Supporting Materials for “Automatic Differentiation to Improve Parameter Estimation for Partially Observed Markov Processes”

Kevin Tan[1], Aaron J. Abkemeier[2], Giles Hooker[1] and Edward L. Ionides[2]
[1]University of Pennsylvania and [2]University of Michigan

Supplementary Content

S1 Proof of Theorem 1 (DMOP-α Functional Forms)

Theorem 1 follows immediately as a consequence of the following results, Lemmas S1 and S2. Here, an A superscript is used to denote ancestral trajectories, so that $x^{A,P,\theta}_{1:n,j}$ contains the history of the prediction particle $x^{P,\theta}_{n,j}$ at observation times $1:n$. Similarly, $g^{A,P,\theta}_{i,j}$ is the ancestral weight of $x^{P,\theta}_{n,j}$ at observation time i.

Lemma S1. *Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the DMOP-α gradient estimate (or, equivalently, the gradient of MOP-α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 1$,*

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J}\sum_{j=1}^J \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y^*_{1:N}|x^{A,F,\theta}_{1:n,j};\theta\big), \tag{S1}$$

yielding the bootstrap filter estimator of Poyiadjis et al. (2011) and Ścibior and Wood (2021).

Proof. Consider the case of MOP-α when $\alpha = 1$ and $\theta = \phi$. We then have a nice telescoping product property for the after-resampling likelihood estimate:

$$\begin{aligned}
\hat{\mathcal{L}}^1(\theta) \quad &:= \quad \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N L_n^\phi \cdot \frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}} = \prod_{n=1}^N L_n^\phi \cdot \frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{F,\theta}_{n-1,j}} \\
&= \quad \left(\frac{1}{J}\sum_{j=1}^J w^{F,\theta}_{N,j}\right)\prod_{n=1}^N L_n^\phi,
\end{aligned} \tag{S2}$$

where the third equality follows from the choice of $\alpha = 1$, and the fourth equality is the resulting telescoping property. The log-derivative identity lets us decompose the score estimate as

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{\nabla_\theta \hat{\mathcal{L}}^1(\theta)}{\hat{\mathcal{L}}^1(\theta)} = \frac{\nabla_\theta\left(\frac{1}{J}\sum_{j=1}^J w^{F,\theta}_{N,j}\right)\prod_{n=1}^N L_n^\phi}{\prod_{n=1}^N L_n^\phi} = \frac{1}{J}\sum_{j=1}^J \nabla_\theta w^{F,\theta}_{N,j}. \tag{S3}$$

From (S3), we see that the derivative of the log-likelihood estimate is

$$\nabla_\theta \hat{\ell}^1(\theta) := \frac{1}{J}\sum_{j=1}^J \nabla_\theta w^{F,\theta}_{N,j}. \tag{S4}$$

We proceed to decompose (S4). First, observe that as $\alpha = 1$,

$$w^{P,\theta}_{n,j} = w^{F,\theta}_{n-1,j}\frac{g^\theta_{n,j}}{g^\phi_{n,j}} = \prod_{i=1}^n \frac{g^{A,P,\theta}_{i,j}}{g^{A,P,\phi}_{i,j}}, \tag{S5}$$

Note that the right hand side of (S5) is the cumulative product of measurement density ratios over the ancestral trajectory of the j-th prediction particle at timestep n. We then use the log-derivative identity again, yielding the following expression for the gradient of the log-weights as the sum of the log measurement densities over the ancestral trajectory,

$$\frac{\nabla_\theta w^{P,\theta}_{n,j}}{w^{P,\theta}_{n,j}} = \nabla_\theta \log w^{P,\theta}_{n,j} \quad = \quad \nabla_\theta \log\left(\prod_{i=1}^n \frac{g^{A,P,\theta}_{i,j}}{g^{A,P,\phi}_{i,j}}\right) \tag{S6}$$

$$= \nabla_\theta \sum_{i=1}^{n} \left(\log g_{i,j}^{A,P,\theta} - \log g_{i,j}^{A,P,\phi} \right) \quad \text{(S7)}$$

$$= \sum_{i=1}^{n} \nabla_\theta \log g_{i,j}^{A,P,\theta}. \quad \text{(S8)}$$

This is equal to the gradient of the logarithm of the conditional density of the observed measurements given the ancestral trajectory of the j-th prediction particle up to timestep n,

$$\begin{aligned}
\nabla_\theta \sum_{n=1}^{N} \log g_{n,j}^{A,P,\theta} &= \nabla_\theta \log \left(\prod_{n=1}^{N} g_{n,j}^{A,P,\theta} \right) && \text{(S9)} \\
&= \nabla_\theta \log \left(\prod_{n=1}^{N} f_{Y_n|X_n}\big(y_n^* | x_{n,j}^{A,P,\theta}; \theta\big) \right) && \text{(S10)} \\
&= \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^* | x_{1:n,j}^{A,P,\theta}; \theta\big). && \text{(S11)}
\end{aligned}$$

Multiplying both sides of the expression by $w_{N,j}^{P,\theta}$ yields an expression for the gradient of the weights at timestep N,

$$\nabla_\theta w_{N,j}^{P,\theta} = w_{N,j}^{P,\theta} \sum_{n=1}^{N} \nabla_\theta \log g_{n,j}^{A,P,\theta} = w_{N,j}^{P,\theta} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^* | x_{1:n,j}^{A,P,\theta}; \theta\big). \quad \text{(S12)}$$

Substituting the above identity into the log-likelihood decomposition obtained earlier in Equation S3 yields

$$\begin{aligned}
\nabla_\theta \hat{\ell}^1(\theta) &= \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta w_{N,j}^{F,\theta} = \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta w_{N,k_j}^{P,\theta} \\
&= \frac{1}{J} \sum_{j=1}^{J} w_{N,k_j}^{P,\theta} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^* | x_{1:n,k_j}^{A,P,\theta}; \theta\big). && \text{(S13)}
\end{aligned}$$

Finally, observing that $\theta = \phi$ implies $w_{N,j}^{F,\theta} = 1$, we obtain

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^* | x_{1:n,j}^{A,F,\theta}; \theta\big). \quad \text{(S14)}$$

This yields the gradient estimators of Poyiadjis et al. (2011); Ścibior and Wood (2021) when applied to the bootstrap filter. □

Note that the variance of the DMOP-α log-likelihood derivative estimate scales poorly with N when $\theta \neq \phi$. This can be seen by observing that

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta w_{N,j}^{F,\theta} = \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta w_{N,k_j}^{P,\theta}$$

$$= \frac{1}{J}\sum_{j=1}^{J} w^{P,\theta}_{N,k_j} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y^*_{1:N}|x^{A,P,\theta}_{1:n,k_j};\theta\big). \tag{S15}$$

When $\theta \neq \phi$, we see that $w^{P,\theta}_{N,k_j} = O(c^N)$. When $\theta = \phi$, this is a special case of the Poyiadjis et al. (2011) estimator, which has $O(N^4)$ variance by a property of functionals of the particle filter (Lemma 2 of Karjalainen et al. (2023) or Proposition 11.3 of Chopin and Papaspiliopoulos (2020)).

Lemma S2. *Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the DMOP-α gradient estimate (or, equivalently, the gradient of MOP-α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 0$,*

$$\nabla_\theta \hat{\ell}^0(\theta) = \frac{1}{J}\sum_{n=1}^{N}\sum_{j=1}^{J} \nabla_\theta \log\Big(f_{Y_n|X_n}\big(y^*_n|x^{F,\theta}_{n,j};\theta\big)\Big), \tag{S16}$$

yielding the estimate of Naesseth et al. (2018) when applied to the bootstrap filter.

Proof. First, write

$$s_{n,j} = \frac{f_{Y_n|X_n}\big(y^*_n|x^{P,\theta}_{n,j};\theta\big)}{f_{Y_n|X_n}\big(y^*_n|x^{P,\phi}_{n,j};\theta\big)} \tag{S17}$$

as shorthand for the measurement density ratios. Observe that, when $\alpha = 0$,, the likelihood estimate becomes

$$\begin{aligned}
\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^{N} L_n^{A,\theta,\alpha} &= \prod_{n=1}^{N} L_n^\phi \cdot \frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}} && \text{(S18)}\\
&= \prod_{n=1}^{N} L_n^\phi \cdot \frac{1}{J}\sum_{j=1}^{J} s_{n,j} && \text{(S19)}\\
&= \prod_{n=1}^{N} L_n^\phi \cdot \frac{1}{J}\sum_{j=1}^{J} \frac{f_{Y_n|X_n}\big(y^*_n|x^{P,\theta}_{n,j};\theta\big)}{f_{Y_n|X_n}\big(y^*_n|x^{P,\phi}_{n,j};\theta\big)}. && \text{(S20)}
\end{aligned}$$

We lose the nice telescoping property observed in the MOP-1 case, but this expression still yields something useful. This is because its gradient when $\theta = \phi$ is therefore

$$\begin{aligned}
\nabla_\theta \hat{\ell}^0(\theta) &:= \sum_{n=1}^{N} \nabla_\theta \log\left(L_n^\phi \frac{1}{J}\sum_{j=1}^{J} s_{n,j}\right) && \text{(S21)}\\
&= \sum_{n=1}^{N} \frac{\nabla_\theta\left(L_n^\phi \frac{1}{J}\sum_{j=1}^J s_{n,j}\right)}{\left(L_n^\phi \frac{1}{J}\sum_{j=1}^J s_{n,j}\right)} && \text{(S22)}\\
&= \sum_{n=1}^{N} \frac{\sum_{j=1}^J \nabla_\theta s_{n,j}}{\sum_{j=1}^J s_{n,j}} && \text{(S23)}\\
&= \sum_{n=1}^{N} \frac{1}{J}\sum_{j=1}^{J} \frac{\nabla_\theta f_{Y_n|X_n}\big(y^*_n|x^{F,\theta}_{n,j};\theta\big)}{f_{Y_n|X_n}(y^*_n|x^{F,\phi}_{n,j};\phi)} && \text{(S24)}
\end{aligned}$$

$$= \frac{1}{J} \sum_{n=1}^{N} \sum_{j=1}^{J} \nabla_\theta \log \left(f_{Y_n|X_n} \left(y_n^* | x_{n,j}^{F,\theta}; \theta \right) \right), \tag{S25}$$

where we use the log-derivative trick in the second equality, observe that $\sum_{j=1}^{J} s_{n,j} = J$ when $\theta = \phi$ in the fourth equality, and use the log-derivative trick again while noting that $\theta = \phi$ in the fifth equality. This yields the desired result. □

S2 Proof of Theorem 2 (Optimization Convergence Analysis)

The analysis in this section is similar to the approach of Roosta-Khorasani and Mahoney (2016), with the caveat that none of the matrix concentration bounds they use apply here as the particles are dependent. We instead use the concentration inequality from Del Moral and Rio (2011) to bound the gradient and Hessian estimates. In this section, we fix $\omega \in \Omega$ only within each filtering iteration, and analyze Algorithm 3 post-iterated filtering.

The convergence analysis in Theorem 2 is limited to the case where $-\ell$ is γ-strongly convex. Under regularity conditions, we expect the log-likelihood to be approximately quadratic in a neighborhood of its maximum, though in practice the likelihood surfaces for POMP models are often highly nonconvex globally. The convergence to an optimum, local or global, must therefore be sensitive to initialization.

For simplicity, we only provide the argument for DMOP-1 and note that the general case follows as the required ϵ-approximation can still be obtained for sufficiently large α. When $\alpha = 1$, the required J is given in the below lemmas. We start by bounding the error on the gradient estimate, in Lemma S3.

Lemma S3 (Concentration of Measure for Gradient Estimate)**.** *Consider the gradient estimate obtained by MOP-1, which we know by Theorem 4 is consistent for the score, where $\theta = \phi$. For $\|\nabla_\theta \hat{\ell}^1(\theta) - \nabla_\theta \ell(\theta)\|_2$ to be bounded by ϵ with probability $1 - \delta$, we require*

$$J > \max \left\{ 2G(\theta) \frac{r_N \sqrt{p}}{\epsilon} \left(1 + h^{-1} \left(\log \left(\frac{2p}{\delta} \right) \right) \right), 8G(\theta)^2 \beta_N^2 \frac{p \log(2p/\delta)}{\epsilon^2} \right\}, \tag{S26}$$

where $NG'(\theta) \leq G(\theta)$ are defined in Assumptions (A3) and (A4), $h(t) = \frac{1}{2}(t - \log(1+t))$, and β_N and r_N are two additional finite model-specific constants that do not depend on J, but do depend on N and p, as defined in Del Moral and Rio (2011). Equivalently, with probability at least $1 - \delta$, it holds that

$$\|\nabla_\theta \hat{\ell}^1(\theta) - \nabla_\theta \ell(\theta)\|_2 \leq G(\theta) \left(\frac{r_N \sqrt{p}}{J} (1 + h^{-1}(\log(2p/\delta))) + \sqrt{\frac{2p \log(2p/\delta)}{J}} \beta_N \right). \tag{S27}$$

Note that, according to Del Moral and Rio (2011), under suitable regularity conditions, r_N and β_N are linear in the trajectory length N. This corresponds to the finding by Poyiadjis et al. (2011) that the variance of the estimate is at least quadratic in the trajectory length, and their remark that the result of Del Moral and Doucet (2003) establishes that the L_p error is bounded by $O(N^2 J^{-1/2})$ (equivalently, the variance is bounded by $O(N^4 J^{-1})$) after accounting for the sum over timesteps. The MOP-1 variance upper bound is therefore in fact $O(N^4)$, in contrast to the MOP-α, where $\alpha < 1$, upper bound of $O(N)$.

Proof. We use the concentration inequality of Del Moral and Rio (2011) to bound the deviation of the gradient estimate from the gradient of the negative log-likelihood in the sup norm with a union bound. Fix $\theta = \phi$. From the decomposition in the proof of Lemma S1, as $w^{F,\theta}_{N,j} = 1$ when $\theta = \phi$, we have that

$$\nabla_\theta \hat{\ell}(\theta) := \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta w^{F,\theta}_{N,j} = \frac{1}{J}\sum_{j=1}^{J}\sum_{n=1}^{N} w^{P,\theta}_{N,k_j} \nabla_\theta \log g^{A,\theta}_{n,k_j} = \frac{1}{J}\sum_{j=1}^{J}\sum_{n=1}^{N} \nabla_\theta \log g^{A,\theta}_{n,k_j}. \tag{S28}$$

Define $\varphi^i_n(x^{F,\theta}_{n,j}) := \frac{\partial}{\partial\theta_i} \log g^{A,\theta}_{n,k_j}$, which is a functional of the filtering particles $x^{F,\theta}_{n,j} = x^{P,\theta}_{n,k_j}$. These are bounded measurable functionals bounded by $G'(\theta)$ by Assumption (A3). Therefore, these have bounded oscillation, satisfying the requirement that $\mathrm{osc}\left(\frac{\partial}{\partial\theta_i}\varphi_i(x^{P,\theta}_{n,j})\right) \leq G'(\theta)$. Note that Del Moral and Rio (2011) in fact assume $\mathrm{osc}(f) \leq 1$, so we simply scale their bound accordingly.

Now we apply the Hoeffding-type concentration inequality in Theorem 1.2 of Del Moral and Rio (2011), and a union bound over each $\varphi^i_n(x^{F,\theta}_{n,j})$, totaling N timesteps and p parameters, to find that

$$\max_{n=1,\dots,N} \left\| \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log g^{A,\theta}_{n,k_j} - \nabla_\theta \ell_n(\theta) \right\|_\infty \leq G'(\theta)\left(\frac{r_N}{J}(1 + h^{-1}(t)) + \sqrt{\frac{2t}{J}}\beta_N \right) \tag{S29}$$

with probability at least $1 - 2Np\exp(-t)$. The concentration inequality justifying (S29) directly concerns the error from the expectation under the filtering distribution. However, we replace this with $\nabla_\theta \ell_n(\theta)$ in (S29) since our regularity conditions ensure that the gradient of the measurement density is bounded and thus, by dominated convergence, the expectation of this gradient equals the gradient of its expectation. Summing over N, and noting that $NG'(\theta) \leq G(\theta)$, we therefore obtain, with the same probability, that

$$\begin{aligned} \left\| \frac{1}{J}\sum_{j=1}^{J}\sum_{n=1}^{N} \nabla_\theta \log g^{A,\theta}_{n,k_j} - \nabla_\theta \ell(\theta) \right\|_\infty &\leq G'(\theta)N\left(\frac{r_N}{J}(1 + h^{-1}(t)) + \sqrt{\frac{2t}{J}}\beta_N \right) &\text{(S30)} \\ &\leq G(\theta)\left(\frac{r_N}{J}(1 + h^{-1}(t)) + \sqrt{\frac{2t}{J}}\beta_N \right). &\text{(S31)} \end{aligned}$$

We split the δ failure probability among these $2Np$ terms, to find $\delta \leq 2Np\exp(-t)$, and therefore, $t \leq \log(2Np/\delta)$, where $h(t) = \frac{1}{2}(t - \log(1+t))$. The two additional model-specific parameters are β_t and r_t, which do not depend on J. The analogous bound for the 2-norm follows from scaling the right-hand side by $\sqrt{p}$, to require

$$\|\nabla_\theta \hat{\ell}(\theta) - \nabla_\theta \ell(\theta)\|_2 \leq G(\theta)\left(\frac{r_N\sqrt{p}}{J}(1 + h^{-1}(\log(2p/\delta))) + \sqrt{\frac{2p\log(2p/\delta)}{J}}\beta_N \right). \tag{S32}$$

We therefore need

$$J > \max\left\{ 2G(\theta)\frac{r_N\sqrt{p}}{\epsilon}\left(1 + h^{-1}\left(\log\left(\frac{2p}{\delta}\right)\right)\right), 8G(\theta)^2\beta_N^2\frac{p\log(2p/\delta)}{\epsilon^2} \right\}. \tag{S33}$$

$\square$

We proceed to study the particle Hessian estimate, providing a guarantee for its positive-definiteness in Lemma S4. This bound is relevant if one chooses to use a second-order method involving a particle Hessian estimate.

Lemma S4 (Minimum Eigenvalue Bound for Hessian Estimate)**.** *Assume that the Hessian of the negative log-likelihood* $H = \sum_{j=1}^{J} \mathbb{E} H_j$ *has a minimum eigenvalue* $0 < \gamma < 1$*, and that* $\mathbb{E}\lambda_{\min}(H_j) = \gamma' > 0$*. If*

$$J > \max\left\{\frac{2r_t(1+h^{-1}(t))+2c}{\gamma'}, \frac{2(2t\beta_t^2+c)^2}{\gamma'^2}\right\} \geq \frac{r_t(1+h^{-1}(t))}{\gamma'} + \sqrt{2tJ}\beta_t/\gamma' + c/\gamma' \tag{S34}$$

then $\hat{H}(\theta)$ *is invertible and positive definite with minimum eigenvalue greater than or equal to* $c \in (0, \sum_{j=1}^{J} \mathbb{E}\lambda_{\min}(H_j))$*, with probability at least* $1 - \exp(-t)$*.*

Proof. Write $\hat{H}(\theta) = \hat{H} = \sum_{j=1}^{J} H_j$ for the estimate of the negative of the Hessian, where H_j is an element of the outer sum over the J particles.

As the negative log-likelihood is convex, we want to bound the minimum eigenvalue of $\hat{H}(\theta)$ from below with high probability, so that all the eigenvalues of $\hat{H}(\theta)$ are positive with high probability. This ensures that the estimated Hessian is invertible and positive-definite.

It is known that the minimum eigenvalue of a symmetric matrix is concave. Therefore, it suffices to show that the first inequality in the below expression

$$0 < \sum_{j=1}^{J} \lambda_{\min}(H_j) \leq \lambda_{\min}\left(\sum_{j=1}^{J} H_j\right) = \lambda_{\min}(\hat{H}) \tag{S35}$$

holds with high probability. We apply the particle Hoeffding concentration inequality from Del Moral and Rio (2011) to find that

$$\frac{1}{J}\sum_{j=1}^{J} \lambda_{\min}(H_j) - \mathbb{E}_{\tilde{\pi}_t}\lambda_{\min}(H_j) = \frac{1}{J}\sum_{j=1}^{J}\lambda_{\min}(H_j) - \gamma' \geq -\frac{r_t}{J}\big(1+h^{-1}(t)\big) - \sqrt{\frac{2t}{J}}\beta_t \tag{S36}$$

$$\sum_{j=1}^{J}\lambda_{\min}(H_j) \geq -r_t\big(1+h^{-1}(t)\big) - \sqrt{2tJ}\beta_t + J\gamma', \tag{S37}$$

with probability at least $1 - \exp(-t)$. Here, $h(t) = \frac{1}{2}(t - \log(1+t))$. The two additional model-specific parameters are β_t and r_t, which do not depend on J.

We additionally require, for $c \in \big(0, \sum_{j=1}^{J} \mathbb{E}\lambda_{\min}(H_j)\big)$,

$$\sum_{j=1}^{J}\lambda_{\min}(H_j) \geq -r_t\big(1+h^{-1}(t)\big) - \sqrt{2tJ}\beta_t + J\gamma' \geq c, \tag{S38}$$

$$J\gamma' \geq c + r_t\big(1+h^{-1}(t)\big) + \sqrt{2tJ}\beta_t. \tag{S39}$$

It is therefore sufficient to have

$$J > \max\left\{\frac{2r_t\big(1+h^{-1}(t)\big)+2c}{\gamma'}, \frac{2(2t\beta_t^2+c)^2}{\gamma'^2}\right\} \geq \frac{r_t\big(1+h^{-1}(t)\big)}{\gamma'} + \sqrt{2tJ}\beta_t/\gamma' + c/\gamma' \tag{S40}$$

for $\hat{H}(\theta)$ to be invertible and positive definite with minimum eigenvalue greater than or equal to c with probability at least $1 - \exp(-t)$.

□

We now proceed with the proof of Theorem 2. In this analysis, we largely follow the proof of Theorem 6 in Roosta-Khorasani and Mahoney (2016). Define $\theta_\eta = \theta_m + \eta p_m$, where $p_m = -(H(\theta_m))^{-1} g(\theta_m)$. As in Roosta-Khorasani and Mahoney (2016), we want to show there is some iteration-independent $\tilde{\eta} > 0$ such that the Armijo condition

$$f(\theta_m + \eta p_m) \leq f(\theta_m) + \eta\beta\, p_m^T\, g(\theta_m), \tag{S41}$$

holds for any $0 < \eta < \tilde{\eta}$ and some $\beta \in (0,1)$. By an argument found in the beginning of the proof of Theorem 6 of Roosta-Khorasani and Mahoney (2016), we have that choosing J such that $\|\nabla_\theta \hat{\ell}(\theta_m) - \nabla_\theta \ell(\theta_m)\| \leq \epsilon$ and $\lambda_{\min}(H(\theta_m)) \geq c > 0$ for each m, yields

$$f(\theta_\eta) - f(\theta_m) \leq \eta p_m^T\, g(\theta_m) + \epsilon\eta\|p_m\| + \eta^2\Gamma\|p_m\|^2/2, \tag{S42}$$

with probability $1 - \delta/2$. From now on, we assume that we are on the success event of this high-probability statement. Consequently, we have

$$p_m^T g(\theta_m) = -p_m^T\, H(\theta_m)\, p_m \geq -c\|p_m\|^2, \tag{S43}$$

and we can obtain a decrease in the objective. Substituting this into the previous expression,

$$f(\theta_\eta) - f(\theta_m) \leq -\eta p_m^T H(\theta_m)\, p_m + \epsilon\eta\|p_m\| + \eta^2\Gamma\|p_m\|^2/2, \tag{S44}$$

the Armijo condition becomes

$$-\eta p_m^T H(\theta_m) p_m + \epsilon\eta\|p_m\| + \eta^2\Gamma\|p_m\|^2/2 \leq \eta\beta p_m^T g(\theta_m) = -\eta\beta p_m^T H(\theta_m) p_m \tag{S45}$$

$$\epsilon\|p_m\| + \eta\Gamma\|p_m\|^2/2 \leq (1-\beta) p_m^T H(\theta_m) p_m \tag{S46}$$

$$\epsilon + \eta\Gamma\|p_m\|/2 \leq c(1-\beta)\|p_m\|. \tag{S47}$$

This holds and guarantees an iteration-independent lower bound if

$$\eta \leq \frac{c(1-\beta)}{\Gamma}, \;\; \epsilon \leq \frac{c(1-\beta)}{2\Gamma}\|g(\theta_m)\| \leq \frac{c(1-\beta)}{2}\|p_m\|, \tag{S48}$$

which is given by our choice of η. Now, first note that

$$\|g(\theta_m)\| - \|\nabla_\theta f(\theta_m)\| \leq \|g(\theta_m) - \nabla_\theta f(\theta_m)\| \leq \epsilon \implies \|\nabla_\theta f(\theta_m)\| \geq \|g(\theta_m)\| - \epsilon \tag{S49}$$

and

$$\|\nabla_\theta f(\theta_m)\| - \|g(\theta_m)\| \leq \|\nabla_\theta f(\theta_m) - g(\theta_m)\| \leq \epsilon \implies \|g(\theta_m)\| \geq \|\nabla_\theta f(\theta_m)\| - \epsilon. \tag{S50}$$

There are now two cases. If the algorithm terminates and $\|g(\theta_m)\| \leq \sigma\epsilon$, we can derive

$$\|\nabla_\theta f(\theta_m)\| \leq \|g(\theta_m)\| + \epsilon = \sigma\epsilon + \epsilon = (\sigma+1)\epsilon. \tag{S51}$$

If the algorithm does not terminate, then $\|g(\theta_m)\| > \sigma\epsilon$. Notice that

$$\epsilon \geq \|g(\theta_m) - \nabla_\theta f(\theta_m)\| \quad \geq \quad \|g(\theta_m)\| - \|\nabla_\theta f(\theta_m)\| \tag{S52}$$

$$
\begin{aligned}
\|\nabla_\theta f(\theta_m)\| + \epsilon &\geq \|g(\theta_m)\| \geq \sigma\epsilon && \text{(S53)}\\
\|\nabla_\theta f(\theta_m)\| &\geq \sigma\epsilon - \epsilon = (\sigma-1)\epsilon && \text{(S54)}\\
\frac{\|\nabla_\theta f(\theta_m)\|}{\sigma-1} &\geq \epsilon, && \text{(S55)}
\end{aligned}
$$

and now

$$
\begin{aligned}
\|\nabla_\theta f(\theta_m)\| - \epsilon &\geq \|\nabla_\theta f(\theta_m)\| - \frac{\|\nabla_\theta f(\theta_m)\|}{\sigma-1} && \text{(S56)}\\
&= \left(1 - \frac{1}{\sigma-1}\right)\|\nabla_\theta f(\theta_m\| && \text{(S57)}\\
&= \frac{\sigma-2}{\sigma-1}\|\nabla_\theta f(\theta_m)\| && \text{(S58)}\\
&\geq \frac{2}{3}\|\nabla_\theta f(\theta_m)\|. && \text{(S59)}
\end{aligned}
$$

Since $\|A^{-1}\| = 1/\sigma_{\min}(A)$,

$$
\begin{aligned}
p_m^T H(\theta_m) p_m &= \big(-(H(\theta_m))^{-1} g(\theta_m)\big)^T H(\theta_m)\big(-(H(\theta_m))^{-1} g(\theta_m)\big) && \text{(S60)}\\
&= g(\theta_m)^T (H(\theta_m))^{-1} g(\theta_m) && \text{(S61)}\\
&\geq \frac{1}{c}\|g(\theta_m)\|^2 && \text{(S62)}\\
&\geq \frac{1}{c}\big(\|\nabla_\theta f(\theta_m)\| - \epsilon\big)^2 && \text{(S63)}\\
&\geq \frac{4}{9c}\|\nabla_\theta f(\theta_m)\|^2. && \text{(S64)}
\end{aligned}
$$

From the assumption that f is γ-strongly convex, $\gamma I \preceq \nabla_\theta^2 - \ell \preceq \Gamma I$, by an implication of γ-strong convexity we have

$$
f(\theta_m) - f(\theta^*) \leq \frac{1}{2\gamma}\big\|\nabla_\theta f(\theta_m)\big\|^2, \tag{S65}
$$

and we put together:

$$
f(\theta_m) - f(\theta^*) \leq \frac{1}{2\gamma}\big\|\nabla_\theta f(\theta_m)\big\|^2 \leq \frac{9c}{4}\frac{1}{2\gamma}\, p_m^T H(\theta_m)\, p_m, \tag{S66}
$$

$$
\frac{8\gamma}{9c}\big(f(\theta_m) - f(\theta^*)\big) \leq \frac{4}{9c}\,\big\|\nabla_\theta f(\theta_m)\big\|^2 \leq p_m^T H(\theta_m)\, p_m, \tag{S67}
$$

$$
f(\theta_m) - f(\theta^*) \leq \frac{9c}{8\gamma}\, p_m^T H(\theta_m)\, p_m, \tag{S68}
$$

$$
-\frac{8\gamma}{9c}\big(f(\theta_m) - f(\theta^*)\big) \geq -\frac{4}{9c}\,\big\|\nabla_\theta f(\theta_m)\big\|^2 \geq -p_m^T H(\theta_m)\, p_m. \tag{S69}
$$

From earlier, as the Armijo condition is fulfilled with our choice of η and ϵ,

$$
\begin{aligned}
f(\theta_{m+1}) - f(\theta_m) &\leq -\eta\, p_m^T H(\theta_m) p_m + \epsilon\eta\|p_m\| + \eta^2\Gamma\|p_m\|^2/2 && \text{(S70)}\\
&\leq -\eta\beta\, p_m^T H(\theta_m)\, p_m && \text{(S71)}
\end{aligned}
$$

$$\leq -\eta\beta \,\frac{8\gamma}{9c}\,\big(f(\theta_m) - f(\theta^*)\big). \tag{S72}$$

Therefore,

$$f(\theta_{m+1}) - f(\theta^*) = f(\theta_{m+1}) - f(\theta_m) + f(\theta_m) - f(\theta^*) \tag{S73}$$

$$\leq f(\theta_m) - f(\theta^*) - \eta\beta \,\frac{8\gamma}{9c}\,\big(f(\theta_m) - f(\theta^*)\big) \tag{S74}$$

$$= \Big(1 - \eta\beta\frac{8\gamma}{9c}\Big)\big(f(\theta_m) - f(\theta^*)\big). \tag{S75}$$

S3 Feynman-Kac Models and Monte Carlo Approximations

In this section, we introduce the Feynman-Kac convention of Del Moral (2004) that has since become commonplace (Karjalainen et al., 2023) for the analysis of the particle filter. The mathematical formalization and notation introduced here will be adopted in the remainder of the analysis, in order to prove Theorems 3, 4, and 5. Let $(\eta_n)_{n=1}^N, (\pi_n)_{n=1}^N, (\rho_n)_{n=1}^N$ be sequences of probability measures on the state space $\mathcal{X}$. This is the sequence of prediction distributions $f_{X_n|Y_{1:n-1}}$, filtering distributions $f_{X_n|Y_{1:n}}$, and posterior distributions $f_{X_{1:n}|Y_{1:n}}$ that we seek to approximate with the particle filter. For any measurable bounded functional h, we adopt the following functional-analytic notation, borrowed from Del Moral (2004); Chopin and Papaspiliopoulos (2020); Karjalainen et al. (2023). We choose our specific choice of notation and definitions to be in line with that of Karjalainen et al. (2023).

Markov kernels and the process model: A Markov kernel M with source $\mathcal{X}_1$ and target $\mathcal{X}_2$ is a map $M : \mathcal{X}_1 \times \mathcal{B}(\mathcal{X}_2) \to [0, 1]$ such that for every set $A \in \mathcal{B}(\mathcal{X}_2)$ and every point $x \in \mathcal{X}_1$, the map $x \mapsto M(x, A)$ is a measurable function of x, and the map $A \mapsto M(x, A)$ is a probability measure on $\mathcal{X}_2$. The quantity $M(x, A)$ can be thought of as the probability of transitioning to the set A given that we are at the point x. If this yields a density, this then corresponds to the process density $f_{X_2|X_1}$ conditional on x and integrated over A.

Markov kernels and measures: For any measure η, any Markov kernel M on $\mathcal{X}$, any point $x \in \mathcal{X}$ and any measurable subset $A \subseteq \mathcal{X}$, let

$$\eta(h) = \int h\,d\eta = \int h(x)\eta(dx), \tag{S76}$$

$$(\eta M)(A) = \int \eta(dx)M(x, A), \tag{S77}$$

$$(Mh)(x) = \int M(x, dy)h(y). \tag{S78}$$

Compositions of Markov kernels: The composition of a Markov kernel M_1 with another Markov kernel M_2 is another Markov kernel, given by

$$(M_1M_2)(x, A) = \int M_1(x, dy)M_2(y, A). \tag{S79}$$

Total variation distance: The total variation distance between two measures μ and ν on $\mathcal{X}$ is

$$\|\mu - \nu\|_{\mathrm{TV}} = \sup_{\|h\|_\infty \leq 1/2} |\mu(\phi) - \nu(\phi)| = \sup_{\mathrm{osc}(h) \leq 1} |\mu(h) - \nu(h)|. \tag{S80}$$

Dobrushin contraction: The Dobrushin contraction coefficient β_{TV} of a Markov kernel M is given by

$$\beta_{\mathrm{TV}}(M) = \sup_{x,y \in \mathcal{X}} \big\|M(x,\cdot) - M(y,\cdot)\big\|_{\mathrm{TV}} = \sup_{\mu,\nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\mathrm{TV}}}{\|\mu - \nu\|_{\mathrm{TV}}}. \tag{S81}$$

Potential functions and the measurement model: A potential function $G : \mathcal{X} \to [0, \infty)$ is a non-negative function of an element of the state space $x \in \mathcal{X}$. In our case, this corresponds to the measurement model, and in our previous notation is written as $g_{n,j} = f_{Y_n|X_n}\big(y_n^* | x_{n,j}^F ; \theta\big) = G_n(x_{n,j}^F)$, where we suppress the dependence on θ for notational simplicity. Note that $G_n(\cdot) = f_{Y_n|X_n}(y_n^* | \cdot\,)$ is the conditional density of the observed measurement at time n, where we condition on the filtering particle $x_{n,j}^F$ as an element of the state space.

Feynman-Kac models: A Feynman-Kac model on $\mathcal{X}$ is a tuple $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$ of an initial probability measure on the state space π_0, a sequence of transition kernels $(M_n)_{n=1}^N$, and a sequence of potential functions $(G_n)_{n=1}^N$. In the notation used in the main text, this corresponds to the starting distribution f_{X_0}, the sequence of transition densities $f_{X_n|X_{n-1}}$, and the measurement densities $f_{Y_n|X_n}$. This induces a set of mappings from the set of probability measures on $\mathcal{X}$ to itself, $\mathcal{P}(\mathcal{X}) \to \mathcal{P}(\mathcal{X})$, as follows:

- The update from the prediction to the filtering distributions is given by

$$\pi_n(dx) = \Psi_n(\eta_n)(dx) = \frac{G_n(x) \cdot \eta_n(dx)}{\eta_n(G_n)}. \tag{S82}$$

- The map from the prediction distribution at timestep n to timestep $n+1$ is given by

$$\Phi_{n+1}(\eta_n) = \Psi_n(\eta_n) M_{n+1}. \tag{S83}$$

- The composition of maps between prediction distributions yields the map from the prediction distribution at time k to the prediction distribution at time n where $k \leq n$,

$$\Phi_{k,n} = \Phi_n \circ \ldots \circ \Phi_{k+1}. \tag{S84}$$

The particle filter: The particle filter then yields a Monte Carlo approximation to the above Feynman-Kac model, via a sequence of mixture Dirac measures. When one resamples at every timestep, the prediction measure at timestep n is then given by

$$\eta_n^J = \frac{1}{J} \sum_{j=1}^J \delta_{x_{n,j}^P}, \tag{S85}$$

and the filtering measure at timestep n is given by

$$\pi_n^J = \frac{\sum_{j=1}^J g_{n,j} \delta_{x_{n,j}^P}}{\sum_{j=1}^J g_{n,j}} \approx \frac{1}{J} \sum_{j=1}^J \delta_{x_{n,j}^F}. \tag{S86}$$

In a slight abuse of notation, we will identify $x^P_{n,1:J} \equiv \eta^J_n$, and $x^F_{n,1:J} \equiv \pi^J_n$. As in Karjalainen et al. (2023), one can view this as an inhomogenous Markov process evolving on $\mathcal{X}^J$. The corresponding Markov transition kernel is then

$$\mathbf{M}_n(x^P_{n-1,1:J}, \cdot) = \left(\Phi_n\left(\eta^J_n\right)\right)^{\otimes J} = \left(\Phi_n\left(\frac{1}{J}\sum_{j=1}^{J}\delta_{x^P_{n,j}}\right)\right)^{\otimes J}, \tag{S87}$$

and the composition of Markov kernels on particles from timestep n to timestep $n+k$ is written

$$\mathbf{M}_{n,n+k} = \mathbf{M}_{n+k} \circ ... \circ \mathbf{M}_n. \tag{S88}$$

One may wonder why Karjalainen et al. (2023) require this process to evolve on $\mathcal{X}^J$. This is because at every timestep n, we in fact draw $X^P_{n,j}|\{X^P_{n-1,1:J} = x^P_{n-1,1:J}\} \sim \mathbf{M}_n(x^P_{n-1,1:J}, \cdot) = \eta_0^{\otimes J}\mathbf{M}_{0,n}$ for $j = 1, ..., J$.

Forgetting of the particle filter: The above formalization yields a result from Karjalainen et al. (2023) on the forgetting of the particle filter that we require for our analysis of the bias, variance, and error of MOP-α. That is, Karjalainen et al. (2023) show that

$$\beta_{\mathrm{TV}}\left(\mathbf{M}_{n,n+k}\right) \leq (1-\epsilon)^{\lfloor k/(O(\log J))\rfloor}, \tag{S89}$$

for some ϵ dependent on $\bar{G}, \underline{\mathrm{G}}, \bar{M}, \underline{\mathrm{M}}$ in Assumptions (A3) and (A2). As a result, the mixing time of the particle filter is only on the order of $O(\log(J))$ timesteps.

Equipped with the above formalisms and results, we are now in a position to provide guarantees on the performance of MOP-α itself.

S4 Proof of Theorem 3 via a Strong Law of Large Numbers for Triangular Arrays of Particles With Off-Parameter Resampling

Here, we prove Theorem 3 by deriving more general result, from which it will be clear that MOP-α targets the filtering distribution and is strongly consistent for the likelihood. We will prove a strong law of large numbers for triangular arrays of particles with off-parameter resampling, meaning that we resample the particles according to an arbitrary resampling rule that is not necessarily in proportion to the target distribution of interest. Using weights that encode the cumulative discrepancy between the resampling distribution and the target distribution (instead of resampling to equal weights, as in the basic particle filter) provides a sufficient correction to ensure almost sure convergence. We now introduce the precise definition of an off-parameter resampled particle filter.

Definition S1 (Off-Parameter Resampled Particle Filters)**.** *An off-parameter resampled particle filter is a Monte Carlo approximation to a Feynman-Kac model* $\left(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N\right)$, *where we inductively define given some* $x^F_{n-1,j}$, $w^F_{n-1,j}$ *comprising the filtering measure approximation* π^J_n*:*

1. *The prediction particles at timestep* n, $x^P_{n,j}$, *are drawn from* $X^P_{n,j} \sim \pi^J_n M_n$, *and the prediction weights are given by* $w^P_{n,j} = w^F_{n-1,j}$.

2. *The prediction measure at timestep n is given by*

$$\eta_n^J = \frac{\sum_{j=1}^J w_{n,j}^P \, \delta_{x_{n,j}^P}}{\sum_{j=1}^J w_{n,j}^P}. \tag{S90}$$

3. *The particles are resampled at every timestep n, yielding indices k_j according to some arbitrary probabilities $(p_{n,j})_{j=1}^J$.*

4. *The filtering measure at timestep n is given by*

$$\pi_n^J = \frac{\sum_{j=1}^J \delta_{x_{n,j}^P} w_{n,j}^P \, g_{n,j}}{\sum_{j=1}^J w_{n,j}^P \, g_{n,j}} \approx \frac{\sum_{j=1}^J \delta_{x_{n,k_j}^P} w_{n,k_j}^P \, g_{n,k_j} / p_{n,k_j}}{\sum_{j=1}^J w_{n,k_j}^P \, g_{n,k_j} / p_{n,k_j}} = \frac{\sum_{j=1}^J w_{n,j}^F \, \delta_{x_{n,j}^F}}{\sum_{j=1}^J w_{n,j}^F}. \tag{S91}$$

5. *The prediction weights at the next timestep are given by $w_{n+1,j}^P = w_{n,j}^F = w_{n,k_j}^P \, g_{n,k_j} / p_{n,k_j}$.*

To prove that the weight correction is sufficient for almost sure convergence, we first introduce what it means for a triangular array of particles to *target* a given target distribution.

Definition S2 (Targeting)**.** *A pair of random vectors (X, W) drawn from some measure g **targets** another measure π if, for any measurable and bounded functional h,*

$$E_g\big[h(X) \cdot W\big] = E_\pi\big[h(X)\big]. \tag{S92}$$

*A particle approximation is a triangular array of pairs of random vectors (X_j^J, W_j^J), $j = 1, 2, \ldots, J$. The particle approximation **targets** π if, for any measurable and bounded functional h,*

$$\frac{\sum_{j=1}^J h(X_j^J) \, W_j^J}{\sum_{j=1}^J W_j^J} \overset{a.s.}{\to} E_\pi\big[h(X)\big] \tag{S93}$$

as $J \to \infty$.

Chopin (2004) asserted without proof that common particle filter algorithms target the filtering distribution in this sense, while Chopin and Papaspiliopoulos (2020) proved a related result assuming bounded densities. We follow a similar approach to Chopin and Papaspiliopoulos (2020), based on showing strong laws of large numbers for triangular arrays, noting that triangular array strong laws do not generally hold without an additional regularity condition such as boundedness.

In order to prove the consistency of our variation on the particle filter, we now present three helper lemmas. The first follows from standard importance sampling arguments, the second from integrating out the marginal, and the third from Bayes' theorem. We state Lemma S5 assuming multinomial resampling, which is convenient for the proof though other resampling strategies may be preferable in practice.

Lemma S5 (Change of Weight Measure)**.** *Suppose that $\{(\tilde{X}_j^J, U_j^J), j = 1, \ldots, J\}$ targets f_X. Now, let $\{(Y_j^J, V_j^J), j = 1, \ldots, J\}$ be a multinomial sample with indices k_j drawn from $\{(\tilde{X}_j^J, U_j^J)\}$ where $(\tilde{X}_j^J, U_j^J)$ is represented, on average, proportional to $\pi_j^J J$ times. Write*

$$(Y_j^J, V_j^J) = \big(\tilde{X}_{k_j}^J, U_{k_j}^J / \pi_{k_j}^J\big). \tag{S94}$$

If the importance sampling weights U_j / π_j are bounded, then $\{(Y_j^J, V_j^J), j = 1, \ldots, J\}$ targets f_X.

Proof. Note that as the Y_j^J are a subsample from X_j^J, h can be a function of Y as well as it is one for X. We then expand

$$\frac{\sum_j h(Y_j^J)\, V_j^J}{\sum_j V_j^J} = \frac{\sum_j h(\tilde{X}_{k_j}^J)\, \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}}. \tag{S95}$$

By hypothesis,

$$\frac{\sum_j h(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[f(X)\big]. \tag{S96}$$

We want to show

$$\frac{\sum_j h(X_{k_j}^J)\, \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}} - \frac{\sum_j h(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} 0. \tag{S97}$$

For this, it is sufficient to show that

$$\sum_j h(\tilde{X}_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J) \frac{U_j^J}{\pi_j^J} \pi_j^J \overset{a.s.}{\to} 0 \tag{S98}$$

since an application of this result with $h(x) = 1$ provides almost sure convergence of the denominator in (S97). Write $g(\tilde{X}_j^J) = h(\tilde{X}_j^J)\frac{U_{k_j}^J}{\pi_{k_j}^J}$. We therefore need to show that

$$\sum_j Z_j^J := \sum_j \left(g(\tilde{X}_{k_j}^J) - g(\tilde{X}_j^J)\, \pi_j^J \right) \overset{a.s.}{\to} 0. \tag{S99}$$

Because the functional h and importance sampling weights $u_{k_j}^J/\pi_{k_j}^J$ are bounded, we have that $\mathbb{E}\big[(Z_j^J)^4\big] < \infty$. We can then follow the argument of Chopin and Papaspiliopoulos (2020) from this point on, where noting that the Z_j^J are conditionally independent given the $\tilde{X}_j^J$ and U_j^J,

$$\begin{aligned} \mathbb{E}\left[\left(\sum_j Z_j^J \right)^4 \middle| (\tilde{X}_j^J, U_j^J)_{j=1}^J \right] &= J\, \mathbb{E}\left[(Z_1^J)^4 | (\tilde{X}_j^J, U_j^J)_{j=1}^J \right] \\ &\quad + 3J(J-1) \left(\mathbb{E}\big[(Z_j^J)^2 \big| (\tilde{X}_j^J, U_j^J)_{j=1}^J\big] \right) &\text{(S100)} \\ &\leq CJ^2, &\text{(S101)} \end{aligned}$$

for some $C > 0$. Taking expectations on both sides yields

$$\mathbb{E}\left[\left(\sum_j Z_j^J \right)^4 \right] \leq CJ^2 \tag{S102}$$

by the tower property. Now by Markov's inequality,

$$\mathbb{P}\left(\left| \frac{1}{J} \sum_{j=1}^J Z_j^J \right| > \epsilon \right) \leq \frac{1}{\epsilon^4 J^4} \mathbb{E}\left[\left(\sum_{j=1}^J Z_j^J \right)^4 \right] \leq \frac{C}{\epsilon^4 J^2}, \tag{S103}$$

and as these terms are summable we can apply Borel-Cantelli to conclude that these deviations happen only finitely often for every $\epsilon > 0$, giving us the almost-sure convergence for

$$\sum_j h(X^J_{k_j}) \frac{u^J_{k_j}}{\pi^J_{k_j}} - \sum_j h(X^J_j) \frac{u^J_j}{\pi^J_j} \pi^J_j \overset{a.s.}{\to} 0. \tag{S104}$$

Similarly, we also have that

$$\sum_j \frac{u^J_j}{\pi^J_j} \pi^J_j - \sum_j \frac{u^J_{k_j}}{\pi^J_{k_j}} \overset{a.s.}{\to} 0, \tag{S105}$$

and the result is proved.

□

Remark: Note that Lemma S5 permits $\pi_{1:J}$ to depend on $\{(X_j, u_j)\}$ as long as the resampling is carried out independently of $\{(X_j, u_j)\}$, conditional on $\pi_{1:J}$.

Lemma S6 (Particle Marginals)**.** *Suppose that $\{(\tilde{X}^J_j, U^J_j), j = 1, \ldots, J\}$ targets f_X. Also suppose that $\tilde{Z}^J_j \sim f_{Z|X}(\cdot|\tilde{X}^J_j)$ where $f_{Z|X}$ is a conditional probability density function corresponding to a joint density $f_{X,Z}$ with marginal densities f_X and f_Z. Then, if the U^J_j are bounded, $\{(\tilde{Z}^J_j, U^J_j)\}$ targets f_Z.*

Proof. We want to show that, for any measurable bounded h,

$$\frac{\sum_j h(\tilde{Z}^J_j)\, U^J_j}{\sum_j U^J_j} \overset{a.s.}{\to} \mathbb{E}_{f_Z}[h(Z)] = \mathbb{E}_{f_X}\big[\mathbb{E}_{f_{Z|X}}[h(Z)|X]\big]. \tag{S106}$$

By assumption, for any measurable and bounded functional g with domain $\mathcal{X}$,

$$\frac{\sum_j g(\tilde{X}^J_j)\, U^J_j}{\sum_j U^J_j} \overset{a.s.}{\to} \mathbb{E}_{f_X}[g(X)]. \tag{S107}$$

Let $\bar{U}^J_j = \frac{J\, U^J_j}{\sum_j U^J_j}$. Examine the numerator and denominator of the quantity

$$\frac{J^{-1} \sum_j h(\tilde{Z}^J_j)\, U^J_j}{J^{-1} \sum_j U^J_j} = \frac{J^{-1} \sum_j h(\tilde{Z}^J_j)\, \bar{U}^J_j}{J^{-1} \sum_j \bar{U}^J_j}. \tag{S108}$$

The denominator converges to 1 almost surely. The numerator, on the other hand, is

$$\frac{1}{J} \sum_j h(\tilde{Z}^J_j)\, \bar{U}^J_j, \tag{S109}$$

and by the same fourth moment argument to the above lemma, it converges almost surely to the limit of its expectation,

$$\lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J} \sum_j h(\tilde{Z}^J_j)\, \bar{U}^J_j\right] \;=\; \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J} \sum_j \mathbb{E}\Big[h(\tilde{Z}^J_j)\, \bar{U}^J_j \Big| \tilde{X}^J_j, \bar{U}^J_j\Big]\right] \tag{S110}$$

$$= \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j \mathbb{E}\Big[h(Z)\Big|X = \tilde{X}_j^J\Big]\,\bar{U}_j^J\right]. \tag{S111}$$

Applying (S107) with $g(x) = \mathbb{E}\big[h(Z)|X = x\big]$, the average on the right hand side converges almost surely to $\mathbb{E}\big\{\mathbb{E}[h(Z)|X]\big\} = \mathbb{E}[h(Z)]$. It remains to swap the limit and expectations. We can do so with the bounded convergence theorem, and therefore obtain

$$\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\,\bar{U}_j^J \overset{a.s.}{\to} \mathbb{E}_{f_Z}[h(Z)]. \tag{S112}$$

□

Lemma S7 (Particle Posteriors)**.** *Suppose that* $\{(X_j^J, U_j^J), j = 1,\dots,J\}$ *targets* f_X*. Also suppose that* $(X_j'^J, U_j'^J) = \big(X_j^J, U_j^J\, f_{Z|X}(z^*|X_j^J)\big)$*. Then, if* $U_j^J\, f_{Z|X}(z^*|X_j^J)$ *and* $U_j^J\, f_{Z|X}(z^*|X_j^J)/f_Z(z^*)$ *are bounded,* $\{(X_j'^J, U_j'^J)\}$ *targets* $f_{X|Z}(\cdot|z^*)$*.*

Proof. Again, we want to show that

$$\frac{\sum_j h(X_j^J)\cdot U_j^J \cdot f_{Z|X}(z^*|X_j^J)}{\sum_j U_j^J \cdot f_{Z|X}(z^*|X_j^J)} \overset{a.s.}{\to} \mathbb{E}_{f_{X|Z}}\big[h(X)\big|z^*\big]. \tag{S113}$$

We already have that for any measurable bounded g,

$$\frac{\sum_j g(X_j^J)\,U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[g(X)\big]. \tag{S114}$$

Consider the following:

$$\frac{J^{-1}\sum_j h(X_j^J)\,f_{Z|X}(z^*|X_j^J)\,U_j^J}{J^{-1}\sum_j U_j^J} \times \left(\frac{J^{-1}\sum_j f_{Z|X}(z^*|X_j^J)\,U_j^J}{J^{-1}\sum_j U_j^J}\right)^{-1}. \tag{S115}$$

We will apply Equation (S114) to the numerator and the denominator in the ratio above individually. The numerator converges to

$$\frac{J^{-1}\sum_j h(X_j^J)\,f_{Z|X}(z^*|X_j^J)\,U_j^J}{J^{-1}\sum_j u_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[h(X)\,f_{Z|X}(z^*|X)\big], \tag{S116}$$

while the reciprocal of the denominator converges to

$$\frac{J^{-1}\sum_j f_{Z|X}(z^*|X_j^J)\,U_j^J}{J^{-1}\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[f_{Z|X}(z^*|X)\big] = f_Z(z^*). \tag{S117}$$

Now we take advantage of the identities

$$\frac{\mathbb{E}_{f_X}\big[h(X)\,f_{Z|X}(z^*|X)\big]}{f_Z(z^*)} = \mathbb{E}_{f_X}\left[\frac{h(X)\,f_{Z|X}(z^*|X)}{f_Z(z^*)}\right] \tag{S118}$$

$$= \mathbb{E}_{f_X}\left[h(X)\frac{f_{X|Z}(X|z^*)}{f_X(X)}\right] = \mathbb{E}_{f_{X|Z}}\big[h(X)|z^*\big], \tag{S119}$$

to give the desired result.

□

Theorem S1 (Off-Parameter Resampled Particle Filters Target the Filtering Distribution)**.** *The off-parameter resampled particle filter as outlined in Definition S1 targets the filtering distribution.*

Proof. Recursively applying Lemmas S5, S6, and S7, we obtain that the off-parameter resampled particle filter targets the posterior. Specifically, suppose inductively that $\{(X^F_{n-1,j}, w^F_{n-1,j})\}$ targets π_{n-1}. Then, Lemma S6 tells us that $\{(X^P_{n,j}, w^P_{n,j})\}$ targets η_n. Lemma S7 tells us that $\{(X^P_{n,j}, w^P_{n,j} g^\theta_{n,j})\}$ therefore targets π_n. Lemma S5 guarantees that the resampling rule, given by

$$\left(X^F_{n,j}, w^F_{n,j}\right) = \left(X^P_{n,k_j}, w^P_{n,k_j}\, g_{n,k_j}/p_{n,k_j}\right), \tag{S120}$$

with resampling probabilities proportional to $p_{n,j}$, therefore also targets π_n. □

Proposition S1 (MOP-1 Targets the Filtering Distribution)**.** *When $\alpha = 1$ or $\phi = \theta$, MOP-α targets the filtering distribution.*

Proof. When $\theta = \phi$, regardless of the value of α, the ratio $\frac{g^\theta_{n,j}}{g^\phi_{n,j}} = 1$, and this reduces to the vanilla particle filter estimate.

When $\alpha = 1$, and $\theta \neq \phi$, the proof is identical to that of S1. Recursively applying Lemmas S5, S6, and S7, we obtain that the MOP-1 filter targets the posterior. Specifically, suppose inductively that $\{(X^{F,\theta}_{n-1,j}, w^{F,\theta}_{n-1,j})\}$ is properly weighted for $f_{X_{n-1}|Y_{1:n-1}}(x_{n-1}|y^*_{1:n-1};\theta)$. Then, Lemma S6 tells us that $\{(X^{P,\theta}_{n,j}, w^{P,\theta}_{n,j})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y^*_{1:n-1};\theta)$. Lemma S7 tells us that $\{(X^{P,\theta}_{n,j}, w^{P,\theta}_{n,j}\, g^\theta_{n,j})\}$ therefore targets $f_{X_n|Y_{1:n}}(x_n|y^*_{1:n};\theta)$. Lemma S5 guarantees that the resampling rule, given by

$$\left(X^{F,\theta}_{n,j}, w^{F,\theta}_{n,j}\right) = \left(X^{P,\theta}_{n,k_j}, w^{P,\theta}_{n,k_j}\, g^\theta_{n,k_j}/g^\phi_{n,k_j}\right), \tag{S121}$$

with resampling weights proportional to $g^\phi_{n,j}$, therefore also targets $f_{X_n|Y_{1:n}}(x_n|y^*_{1:n};\theta)$. □

This has addressed filtering, but not quite yet the likelihood evaluation. For this we use the following lemma.

Lemma S8 (Likelihood Proper Weighting)**.** *$f_{Y_n|Y_{1:n-1}}(y^*_n|y^*_{1_n-1};\theta)$ is strongly consistently estimated by either the before-resampling estimate,*

$$L^{B,\theta}_n = \frac{\sum_{j=1}^J g^\theta_{n,j}\, w^{P,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}}, \tag{S122}$$

or by the after-resampling estimate,

$$L^{A,\theta}_n = L^\phi_n \frac{\sum_{j=1}^J w^{F,\theta}_{n,j}}{\sum_{j=1}^J w^{P,\theta}_{n,j}}. \tag{S123}$$

where L^ϕ_n is as defined in the various algorithms.

Here, (S122) is a direct consequence of our earlier result that $\{(X^{P,\theta}_{n,j}, w^{P,\theta}_{n,j})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y^*_{1:n-1};\theta)$. To see (S123), we write the numerator of (S5) as

$$L^\phi_n \sum_{j=1}^J \left[\frac{g^\theta_{n,j}}{g^\phi_{n,j}}\, w^{P,\theta}_{n,j}\right] \frac{g^\phi_{n,j}}{L^\phi_n} = L^\phi_n \sum_{j=1}^J w^{FC,\theta}_{n,j}\, \frac{g^\phi_{n,j}}{L^\phi_n}. \tag{S124}$$

Using Lemma S5, we resample according to probabilities $\frac{g^{\phi}_{n,j}}{L^{\phi}_n}$ to see this is properly estimated by

$$L^{\phi}_n \sum_{j=1}^{J} w^{F,\theta}_{n,j}, \tag{S125}$$

from which we obtain (S123). Using Lemma S8, we obtain a likelihood estimate,

$$L^{A,\theta} = \prod_{n=1}^{N} \left(L^{\phi}_n \frac{\sum_{j=1}^{J} w^{F,\theta}_{n,j}}{\sum_{j=1}^{J} w^{P,\theta}_{n,j}} \right). \tag{S126}$$

Since $w^{F,\theta}_{n,j} = w^{P,\theta}_{n+1,j}$, this is a telescoping product. The remaining terms are $\sum_{j=1}^{J} w^{P,\theta}_{0,j} = J$ on the denominator and $\sum_{j=1}^{J} w^{F,\theta}_{N,j}$ on the numerator. This derives the MOP-1 likelihood estimates.

$L^{B,\theta}$ should generally be preferred in practice, since there is no reason to include the extra variability from resampling when calculating the conditional log likelihood, but it lacks the nice telescoping product that lets us derive exact expressions for the gradient in Theorem 1.

S5 Proof of Theorem 4 via Consistency of Off-Parameter Resampled Gradient Estimates

We now provide a result showing that under sufficient regularity conditions, one can interchange the order of differentiation and expectation of functionals of off-parameter resampled particle estimates, showing that if an off-parameter resampled particle estimate is consistent for some estimand, its derivative is consistent for the derivative of the estimand as well.

One may wonder why this may be necessary, given that we have already shown strong consistency for measurable bounded functionals in Theorem S1. If we require the gradient to be bounded as well, then the gradient is also a bounded functional. The answer is that the strong consistency is for expectations under the filtering distribution, so Theorem S1 only establishes strong consistency of the gradient of the estimate to its expectation under the filtering distribution, $\nabla_\theta \eta^J_n(h_\theta) \overset{a.s.}{\to} \eta_n(\nabla_\theta h_\theta)$, which is not a-priori the gradient of the estimand $\nabla_\theta \eta_n(h_\theta)$. We require an interchange of the derivative and expectation, which we show is possible when the particle estimates have two continuous uniformly bounded derivatives over all J below.

Theorem S2 (Off-Parameter Particle Filters Yield Strongly Consistent Estimates of Derivatives of Functionals)**.** *Let $h_\theta : \mathcal{X} \to \mathbb{R}$ be a measurable bounded functional of particles, where $\eta^J_n(h_\theta)$ has two continuous derivatives uniformly bounded over all J by H^* for almost every $\omega \in \Omega$ and $\theta \in \Theta$. If it holds that $\eta^J_n(h_\theta) \overset{a.s.}{\to} \eta(h_\theta) = h^*_\theta$, then we also have that $\nabla_\theta \eta^J_n(h_\theta) \overset{a.s.}{\to} \eta_n(\nabla_\theta h_\theta) = \nabla_\theta \eta_n(h_\theta) = \nabla_\theta h^*_\theta$.*

Proof. Fix $\omega \in \Omega$. The sequence $(\nabla_\theta \eta^J_n(h_\theta)(\omega))_{J\in\mathbb{N}}$ is uniformly bounded over all J, by assumption. The sequence is also uniformly equicontinuous. To see this, by assumption, the second derivative of $\eta^J_n(h_\theta)(\omega)|_{\theta=\theta'}$ is also uniformly bounded over all J for almost every $\omega \in \Omega$ and every $\theta' \in \Theta$. A set of functions with derivatives bounded by the same constant is uniformly Lipschitz, and therefore uniformly equicontinuous. So the sequence $(\eta^J_n(h_\theta)\omega))_{J\in\mathbb{N}}$ is uniformly equicontinuous over θ for almost every $\omega \in \Omega$. Explicitly, for almost every $\omega \in \Omega$ and every $\epsilon > 0$, there exists some $\delta(\omega) > 0$ such that for every $||\theta - \theta'||_\infty < \delta$ and every $J \in \mathbb{N}$ we have that

$$\big|| \eta_n^J(h_\theta)(\omega) - \eta_n^J(h_{\theta'})(\omega)\big||_\infty < \epsilon. \tag{S127}$$

Then, by Arzela-Ascoli, there exists a uniformly convergent subsequence. We claim that there is only one subsequential limit. When the gradient is bounded, we can treat the gradient as a bounded functional. So by Theorem S1 the sequence $(\eta_n^J(h_\theta)(\omega))_{J\in\mathbb{N}}$ converges pointwise for $\theta = \phi$ and almost every $\omega \in \Omega$, and there is therefore only one subsequential limit. The sequence therefore converges uniformly to its limit $\lim_{J\to\infty} \eta_n^J(h_\theta)(\omega)$. Therefore, with uniform convergence for the derivatives established, we can swap the limit and derivative, and obtain that for almost every $\omega \in \Omega$,

$$\lim_{J\to\infty} \eta_n^J(h_\theta)(\omega) = \nabla_\theta \lim_{J\to\infty} \eta_n^J(h_\theta)(\omega). \tag{S128}$$

Again from Theorem S1, we know that

$$\eta_n^J(h_\theta)(\omega) \to \eta_n(h_\theta) = h_\theta^* \tag{S129}$$

for almost every $\omega \in \Omega$. We then have that for almost every $\omega \in \Omega$,

$$\lim_{J\to\infty} \eta_n^J(h_\theta)(\omega) = \nabla_\theta \eta_n(h_\theta) = \nabla_\theta h_\theta^*, \tag{S130}$$

as we wanted. □

Theorem 4 is proved as a corollary of Theorem S2, where we apply $\eta_n^J(h_\theta) = \hat{\mathcal{L}}_J^1(\theta)$, $\eta_n(h_\theta) = \mathcal{L}(\theta) = h_\theta^*$, and then use the continuous mapping theorem.

S6 Proof of Theorem 5 (Bias-Variance Analysis)

In this section, we prove various rates on the bias, variance, and MSE of DMOP-α. First, we note the following relation between the Dobrushin contraction coefficient and the alpha-mixing coefficients in our context below.

Lemma S9. *Setting X to be the particle collection at time m, and Y at time n, we have that the alpha mixing coefficients,*

$$\int \big|f_{XY}(x,y) - f_X(x)\, f_Y(y)\big|\, dx\, dy < \alpha, \tag{S131}$$

are bounded by the Dobrushin coefficient, i.e we have

$$\int \big|f_{Y|X}(y|x_1) - f_{Y|X}(y|x_2)\big|\, dy < \alpha \tag{S132}$$

for all x_1, x_2.

Proof. We rewrite the alpha-mixing assertion as

$$\int\int \big|f_{Y|X}(y|x) - f_Y(y)\big|\, dy\, f_X(x)\, dx. \tag{S133}$$

We claim that the Dobrushin coefficient implies

$$\int \left| f_{Y|X}(y|x_1) - f_Y(y) \right| dy < \alpha \tag{S134}$$

for all x_1. This is shown as follows:

$$\begin{aligned}
\int \left| f_{Y|X}(y|x) - f_Y(y) \right| dy &= \int \left| \int \left[f_{Y|X}(y|x_1) - f_{Y|X}(y|x) \right] f_X(x)\, dx \right| dy && \text{(S135)}\\
&< \int \int \left| f_{Y|X}(y|x_1) - f_{Y|X}(y|x) \right| dy\, f_X(x)\, dx && \text{(S136)}\\
&< \int \alpha f_X(x)\, dx && \text{(S137)}\\
&= \alpha && \text{(S138)}
\end{aligned}$$

We then have the desired result. □

As a warm-up, we consider a variance bound for DMOP-0. Note that we made Assumptions (A2) and (A3) to leverage the results from Karjalainen et al. (2023) on the forgetting of the particle filter. This is required to show error bounds on the gradient estimates that we provide, namely that the error of the DMOP-1 estimator, corresponding to the estimator of Poyiadjis et al. (2011), is $O(N^4)$, and that the variance of DMOP-α for any $\alpha < 1$ is $O(N)$. We start by decomposing the MOP-α estimator as follows, for the case $\theta = \phi$,

$$\hat{\mathcal{L}}(\theta) := \prod_{n=1}^{N} L_n^{A,\theta,\alpha} = \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta}} = \prod_{n=1}^{N} L_n^{A,\theta,\alpha} = \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} (w_{n-1,j}^{F,\theta})^{\alpha}}. \tag{S139}$$

Now, to illustrate the proof strategy for the general case in an easier context, we first analyze the special case when $\alpha = 0$. We now prove the variance bound for this case, presented below. To our knowledge, this result for the variance of the gradient estimate of Naesseth et al. (2018) is novel.

Theorem S3 (Variance of MOP-0 Gradient Estimate)**.** *The variance of DMOP-*0*, i.e. the algorithm of Naesseth et al. (2018), is*

$$\mathrm{Var}\big(\nabla_\theta \hat{\ell}^0(\theta)\big) \lesssim \frac{Np\, G'(\theta)^2}{J}. \tag{S140}$$

Proof. When $\alpha = 0$, we have

$$\begin{aligned}
\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^{N} L_n^{A,\theta,0} &= \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta,0}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta,0}} && \text{(S141)}\\
&= \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{1}{J} \sum_{j=1}^{J} s_{n,j} = \prod_{n=1}^{N} L_n^{\phi} \cdot \frac{1}{J} \sum_{j=1}^{J} \frac{f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{P,\theta};\theta\big)}{f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{P,\phi};\phi\big)}. && \text{(S142)}
\end{aligned}$$

Similarly to the proof of Lemma S2, we define

$$s_{n,j} = \frac{f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{P,\theta};\theta\big)}{f_{Y_n|X_n}\big(y_n^*|x_{n,j}^{P,\phi};\phi\big)}, \tag{S143}$$

and from the exact same arguments conclude that its gradient when $\theta = \phi$ is therefore

$$\nabla_\theta \hat{\ell}^0(\theta) := \sum_{n=1}^{N} \nabla_\theta \log \left(L_n^\phi \frac{1}{J} \sum_{j=1}^{J} s_{n,j} \right) \tag{S144}$$

$$= \sum_{n=1}^{N} \frac{\nabla_\theta \left(L_n^\phi \frac{1}{J} \sum_{j=1}^{J} s_{n,j} \right)}{\left(L_n^\phi \frac{1}{J} \sum_{j=1}^{J} s_{n,j} \right)} \tag{S145}$$

$$= \sum_{n=1}^{N} \frac{\sum_{j=1}^{J} \nabla_\theta s_{n,j}}{\sum_{j=1}^{J} s_{n,j}} \tag{S146}$$

$$= \sum_{n=1}^{N} \frac{1}{J} \sum_{j=1}^{J} \frac{\nabla_\theta f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\phi}; \phi)} \tag{S147}$$

$$= \frac{1}{J} \sum_{n=1}^{N} \sum_{j=1}^{J} \nabla_\theta \log \left(f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta) \right), \tag{S148}$$

where we use the derivative of the logarithm, observe that $\sum_{j=1}^{J} s_{n,j} = J$ when $\theta = \phi$, and use the derivative of the logarithm where $\theta = \phi$ again. We do this to establish that the gradient of the log-likelihood estimate is given by the sum of terms over all N and J:

$$\nabla_\theta \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^{N} \sum_{j=1}^{J} \nabla_\theta \log \left(f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta) \right). \tag{S149}$$

Therefore, for a given θ_i in the parameter vector θ,

$$\mathrm{Var}\left(\frac{\partial}{\partial_{\theta_i}} \hat{\ell}^0(\theta) \right) = \frac{1}{J^2} \mathrm{Var}\left(\sum_{n=1}^{N} \sum_{j=1}^{J} \frac{\partial}{\partial_{\theta_i}} \log \left(f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta) \right) \right) \tag{S150}$$

$$= \frac{1}{J^2} \sum_{n=1}^{N} \mathrm{Var}\left(\sum_{j=1}^{J} \frac{\partial}{\partial_{\theta_i}} \log \left(f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta) \right) \right)$$

$$+ 2 \sum_{m<n} \mathrm{Cov}\left(\frac{1}{J} \sum_{j=1}^{J} \frac{\partial}{\partial_{\theta_i}} \log \left(f_{Y_m|X_m}(y_m^* | x_{m,j}^{F,\theta}; \theta) \right), \frac{1}{J} \sum_{j=1}^{J} \frac{\partial}{\partial_{\theta_i}} \log \left(f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta) \right) \right)$$

$$= \sum_{n=1}^{N} \mathrm{Var}\left(\frac{\partial}{\partial_{\theta_i}} \hat{\ell}_n^0(\theta) \right) + 2 \sum_{m<n} \mathrm{Cov}\left(\frac{\partial}{\partial_{\theta_i}} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial_{\theta_i}} \hat{\ell}_n^0(\theta) \right). \tag{S151}$$

Here, we use Assumptions (A2) and (A3) that ensure strong mixing. We know from Theorem 3 of Karjalainen et al. (2023) that when $\mathbf{M}_{n,n+k}$ is the k-step Markov operator from timestep n and $\beta_{\mathrm{TV}}(M) = \sup_{x,y \in E} \|M(x,\cdot) - M(y,\cdot)\|_{\mathrm{TV}} = \sup_{\mu,\nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\mathrm{TV}}}{\|\mu - \nu\|_{\mathrm{TV}}}$ is the Dobrushin contraction coefficient of a Markov operator,

$$\beta_{\mathrm{TV}}(\mathbf{M}_{n,n+k}) \leq (1-\epsilon)^{\lfloor k/(c \log(J)) \rfloor}, \tag{S152}$$

i.e. the mixing time of the particle filter is $O(\log(J))$, where ϵ and c depend on $\bar{M}, \underline{\mathrm{M}}, \bar{G}, \underline{\mathrm{G}}$ in (A2) and (A3).

By Lemma S9, the particle filter itself is strong mixing, with α-mixing coefficients $a(k) \leq (1-\epsilon)^{\lfloor k/(c\log(J))\rfloor}$. Therefore, functions of particles are strongly mixing as well, with α-mixing coefficients bounded by the original (to see this, observe that the σ-algebra of the functionals is contained within the original σ-algebra). Therefore, by Davydov's inequality, noting that $\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta) \leq G'(\theta)$ by Assumption (A3), and without loss of generality labeling m and n such that $\mathbb{E}\big[(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_m(\theta))^4\big]^{1/4} \leq \mathbb{E}\big[(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta))^4\big]^{1/4}$, we see that

$$\operatorname{Cov}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_m(\theta), \frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right) \leq a(n-m)^{1/2}\,\mathbb{E}\left[\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_m(\theta)\right)^4\right]^{1/4}\mathbb{E}\left[\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right)^4\right]^{1/4} \tag{S153}$$

$$\leq a(n-m)^{1/2}\,\mathbb{E}\left[\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right)^4\right]^{1/2}. \tag{S154}$$

To bound this, we use the fact that

$$\mathbb{E}\left[\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right)^4\right] = \mathbb{E}\left[\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right)^4 - \mathbb{E}\left[\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right)^2\right]^2\right] + \mathbb{E}\left[\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right)^2\right]^2 \tag{S155}$$

alongside Lemma 2 of Karjalainen et al. (2023), which shows that

$$\mathbb{E}\left[\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right)^4 - \mathbb{E}\left[\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right)^2\right]^2\right] \lesssim \frac{G'(\theta)^4}{J^2}, \tag{S156}$$

$$\mathbb{E}\left[\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta) - \mathbb{E}\left[\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right]\right)^2\right] \lesssim \frac{G'(\theta)^2}{J}. \tag{S157}$$

It follows that

$$\mathbb{E}\left[\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right)^4\right]^{1/2} \lesssim \sqrt{\frac{G'(\theta)^4}{J^2} + \left(\frac{G'(\theta)^2}{J}\right)^2} = \frac{G'(\theta)^2}{J}, \tag{S158}$$

and we conclude that

$$\operatorname{Cov}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_m(\theta), \frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right) \leq (1-\epsilon)^{\frac{1}{2}\lfloor\frac{|n-m|}{c\log(J)}\rfloor}\frac{G'(\theta)^2}{J}. \tag{S159}$$

Putting it all together, we see that

$$\operatorname{Var}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^\alpha(\theta)\right) = \sum_{n=1}^N \operatorname{Var}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right) + 2\sum_{m<n}\operatorname{Cov}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_m(\theta), \frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right) \tag{S160}$$

$$\leq \frac{NG'(\theta)^2}{J} + 2\sum_{n=1}^{N-1}(N-n)(1-\epsilon)^{\frac{1}{2}\lfloor\frac{n}{c\log(J)}\rfloor}\frac{G'(\theta)^2}{J} \tag{S161}$$

$$\leq \frac{G'(\theta)^2}{J}\left(N + 2\sum_{n=1}^{N-1}(N-n)(1-\epsilon)^{\frac{1}{2}\lfloor\frac{n}{c\log(J)}\rfloor}\right) \tag{S162}$$

$$\lesssim \frac{G'(\theta)^2 N}{J}. \tag{S163}$$

It then follows that as $\theta \in \Theta \subseteq \mathbb{R}^p$,

$$\mathrm{Var}\big(\nabla_\theta \hat{\ell}^\alpha(\theta)\big) \lesssim \frac{Np\, G'(\theta)^2}{J}. \tag{S164}$$

□

Note that the factor of N that pops up here is due to the use of the unnormalized gradient. If one divides the gradient estimate by $\sqrt{N}$ before usage, the variance does not depend on the horizon. If one divides by N, the error is $O(1/\sqrt{NJ})$.

The variance here of $\frac{Np\, G'(\theta)^2}{J}$ can be thought of as a lower bound for the variance of MOP-α. We will later see that the variance of MOP-α contains this term, and an additional term that corresponds to the memory of the MOP-α gradient estimate.

Having obtained a variance bound for DMOP-0, we are now in a position to tackle a variance bound for DMOP-α. We analyze the case for $\alpha \in (0,1)$, using the following proof strategy. Instead of directly analyzing the variance of the MOP-α gradient estimate, we will analyze the variance of a modified estimator where we truncate the weights at k timesteps, so this estimator only looks at most k timesteps back while accumulating the importance weights. We write

$$\hat{s}_k^\alpha(\theta) := \sum_{n=1}^N \log\left(\frac{\sum_{j=1}^J \prod_{i=n-k}^{n} \left(\frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}\right)^{\alpha^{(n-i)}}}{\sum_{j=1}^J \prod_{i=n-k}^{n-1} \left(\frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}\right)^{\alpha^{(n-i)}}} \right) \tag{S165}$$

for the log-likelihood estimate from the k-truncated estimator with discount factor α. It is the sum of components

$$\hat{s}_{n,k}^\alpha(\theta) := \frac{1}{J}\sum_{j=1}^J \sum_{i=n-k}^{n} \left(\alpha^{n-i} \log\left(g_{i,j}^{A,F,\theta}\right) - \alpha^{n-i+1} \nabla_\theta \log\left(g_{i-1,j}^{A,F,\theta}\right)\right) \tag{S166}$$

over timesteps N.

This enables us to establish strong mixing for the truncated estimator and its gradient, in order to get a bound on the variance that is $O(NJ^{-1})$. We emphasize that this truncated estimator is only a theoretical construct – we do not actually use the truncated estimator. The truncated estimator, as we note in the discussion, possesses similar theoretical guarantees as the MOP-α estimator, but would require the user to specify the number of timesteps to truncate at instead of a discount factor.

Coupled with a bound ensuring that the k-truncated estimator provides an estimate that is close to that of MOP-α proper, we get a bound on the variance of MOP-α that comprises of the variance of the k-truncated estimator plus the error, for any $k \leq N$. It then holds that the final variance bound on MOP-α is given by the minimum over all k, and is never larger than $O(\psi(\alpha) + NJ^{-1})$ for some function ψ increasing in α.

Theorem S4 (Variance of DMOP-α)**.** *When $\alpha \in (0,1)$, the variance of the gradient estimate from MOP-α is*

$$\mathrm{Var}\big(\nabla_\theta \hat{\ell}^\alpha(\theta)\big) \lesssim \min_{k \leq N} \left(\frac{k^2 G'(\theta)^2 Np}{(1-\alpha)^2 J} + \frac{\alpha^{2k}}{(1-\alpha)^2} Np\, G'(\theta)^2 \right). \tag{S167}$$

Proof. Using the derivative of the logarithm and that $w_{n,j}^{F,\theta,\alpha} = w_{n,j}^{F,\theta,\alpha,k} = 1$ when $\theta = \phi$,

$$\left\|\nabla_\theta \log\left(\sum_{j=1}^{J} w_{n,j}^{F,\theta,\alpha}\right) - \nabla_\theta \log\left(\sum_{j=1}^{J} w_{n,j}^{F,\theta,\alpha,k}\right)\right\|_2 \tag{S168}$$

$$= \left\|\frac{\nabla_\theta \sum_{j=1}^{J} w_{n,j}^{F,\theta,\alpha}}{\sum_{j=1}^{J} w_{n,j}^{F,\theta,\alpha}} - \frac{\nabla_\theta \sum_{j=1}^{J} w_{n,j}^{F,\theta,\alpha,k}}{\sum_{j=1}^{J} w_{n,j}^{F,\theta,\alpha,k}}\right\|_2 \tag{S169}$$

$$= \left\|\frac{1}{J}\sum_{j=1}^{J} \nabla_\theta w_{n,j}^{F,\theta,\alpha} - \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta w_{n,j}^{F,\theta,\alpha,k}\right\|_2 \tag{S170}$$

$$= \left\|\frac{1}{J}\sum_{j=1}^{J} \frac{\nabla_\theta w_{n,j}^{F,\theta,\alpha}}{w_{n,j}^{F,\theta,\alpha}} - \frac{1}{J}\sum_{j=1}^{J} \frac{\nabla_\theta w_{n,j}^{F,\theta,\alpha,k}}{w_{n,j}^{F,\theta,\alpha,k}}\right\|_2 \tag{S171}$$

$$= \left\|\frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log\left(w_{n,j}^{F,\theta,\alpha}\right) - \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log\left(w_{n,j}^{F,\theta,\alpha,k}\right)\right\|_2. \tag{S172}$$

This lets us bound the cumulative weight discrepancies by

$$\left\|\nabla_\theta \log\left(\sum_{j=1}^{J} w_{n,j}^{F,\theta,\alpha}\right) - \nabla_\theta \log\left(\sum_{j=1}^{J} w_{n,j}^{F,\theta,\alpha,k}\right)\right\|_2$$

$$= \left\|\frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log\left(w_{n,j}^{F,\theta,\alpha}\right) - \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log\left(w_{n,j}^{F,\theta,\alpha,k}\right)\right\|_2 \tag{S173}$$

$$= \left\|\frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \left(\log\left(w_{n,j}^{F,\theta,\alpha}\right) - \log\left(w_{n,j}^{F,\theta,\alpha,k}\right)\right)\right\|_2 \tag{S174}$$

$$= \left\|\frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \left(\sum_{i=1}^{n} \alpha^{(n-i)} \log\left(\frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}\right) - \sum_{i=n-k}^{n} \alpha^{(n-i)} \log\left(\frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}\right)\right)\right\|_2 \tag{S175}$$

$$= \left\|\frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \left(\sum_{i=1}^{n-k} \alpha^{(n-i)} \log\left(\frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}\right)\right)\right\|_2 \tag{S176}$$

$$\leq \frac{1}{J}\sum_{j=1}^{J}\sum_{i=1}^{n-k} \alpha^{(n-i)} \left\|\nabla_\theta \log\left(g_{i,j}^{A,F,\theta}\right)\right\|_2 \tag{S177}$$

$$\leq \frac{1}{J}\sum_{j=1}^{J}\sum_{i=1}^{n-k} \alpha^{(n-i)} p\, G'(\theta)^2 \tag{S178}$$

$$\leq \sqrt{p}\, G'(\theta) \frac{\alpha^k - \alpha^n}{1-\alpha}, \tag{S179}$$

where the second-last line follows from Assumption (A3).

Now, we bound the variance of $\nabla_\theta \hat{s}_k^\alpha(\theta)$. Recall that we defined

$$\nabla_\theta \hat{s}_k^\alpha(\theta) = \sum_{n=1}^{N} \nabla_\theta \log \left(\frac{\sum_{j=1}^J \prod_{i=n-k}^{n} \left(\frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}} \right)^{\alpha^{(n-i)}}}{\sum_{j=1}^J \prod_{i=n-k}^{n-1} \left(\frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}} \right)^{\alpha^{(n-i)}}} \right) \tag{S180}$$

$$= \sum_{n=1}^{N} \nabla_\theta \left(\log\left(\sum_{j=1}^J w_{n,j}^{F,\theta,\alpha,k} \right) - \log\left(\sum_{j=1}^J w_{n-1,j}^{A,F,\theta,\alpha,k} \right) \right). \tag{S181}$$

We can then decompose this expression with the derivative of the logarithm while noting that $w_{n,j}^{F,\theta,\alpha,k} = 1$ whenever $\theta = \phi$, to see that

$$\nabla_\theta \hat{s}_k^\alpha(\theta) = \sum_{n=1}^{N} \nabla_\theta \left(\log\left(\sum_{j=1}^J w_{n,j}^{F,\theta,\alpha,k} \right) - \log\left(\sum_{j=1}^J w_{n-1,j}^{A,F,\theta,\alpha,k} \right) \right) \tag{S182}$$

$$= \sum_{n=1}^{N} \left(\frac{\sum_{j=1}^J \nabla_\theta w_{n,j}^{F,\theta,\alpha,k}}{\sum_{j=1}^J w_{n,j}^{F,\theta,\alpha,k}} - \frac{\sum_{j=1}^J \nabla_\theta w_{n-1,j}^{A,F,\theta,\alpha,k}}{\sum_{j=1}^J w_{n-1,j}^{A,F,\theta,\alpha,k}} \right) \tag{S183}$$

$$= \sum_{n=1}^{N} \left(\frac{1}{J} \sum_{j=1}^J \nabla_\theta w_{n,j}^{F,\theta,\alpha,k} - \frac{1}{J} \sum_{j=1}^J \nabla_\theta w_{n-1,j}^{A,F,\theta,\alpha,k} \right) \tag{S184}$$

$$= \frac{1}{J} \sum_{n=1}^{N} \sum_{j=1}^J \nabla_\theta \left(w_{n,j}^{F,\theta,\alpha,k} - w_{n-1,j}^{A,F,\theta,\alpha,k} \right). \tag{S185}$$

It now follows that we need to bound the variance at a single timestep, namely

$$\operatorname{Var}\left(\frac{1}{J} \sum_{j=1}^J \nabla_\theta \left(w_{n,j}^{F,\theta,\alpha,k} - w_{n-1,j}^{A,F,\theta,\alpha,k} \right) \right). \tag{S186}$$

We use the derivative of the logarithm yet again to find, noting that $w_{n,j}^{F,\theta,\alpha,k} = 1$ when $\theta = \phi$, that

$$\nabla_\theta w_{n,j}^{F,\theta,\alpha,k} = \frac{\nabla_\theta w_{n,j}^{F,\theta,\alpha,k}}{w_{n,j}^{F,\theta,\alpha,k}} = \nabla_\theta \log(w_{n,j}^{F,\theta,\alpha,k}) = \sum_{i=n-k}^{n} \alpha^{n-i} \nabla_\theta \log\left(\frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}} \right) \tag{S187}$$

$$= \sum_{i=n-k}^{n} \alpha^{n-i} \nabla_\theta \log\left(g_{i,j}^{A,F,\theta} \right). \tag{S188}$$

Then,

$$\operatorname{Var}\left(\nabla_\theta \big(w_{n,j}^{F,\theta,\alpha,k} - w_{n-1,j}^{A,F,\theta,\alpha,k} \big) \right) \tag{S189}$$

$$= \operatorname{Var}\left(\nabla_\theta \left(\sum_{i=n-k}^{n} \alpha^{n-i} \nabla_\theta \log\left(g_{i,j}^{A,F,\theta} \right) - \sum_{i=n-k-1}^{n-1} \alpha^{n-i} \nabla_\theta \log\left(g_{i,j}^{A,F,\theta} \right) \right) \right) \tag{S190}$$

$$= \mathrm{Var}\left(\sum_{i=n-k}^{n} \nabla_\theta \left(\alpha^{n-i}\log\left(g_{i,j}^{A,F,\theta}\right) - \alpha^{n-i+1}\log\left(g_{i-1,j}^{A,F,\theta}\right)\right)\right). \tag{S191}$$

Note that

$$\nabla_\theta \hat{s}_{n,k}^{\alpha}(\theta) := \frac{1}{J}\sum_{j=1}^{J}\sum_{i=n-k}^{n} \nabla_\theta \left(\alpha^{n-i}\log\left(g_{i,j}^{A,F,\theta}\right) - \alpha^{n-i+1}\nabla_\theta \log\left(g_{i-1,j}^{A,F,\theta}\right)\right) \tag{S192}$$

is a function bounded by $C\frac{G'(\theta)(1-\alpha^k)}{1-\alpha} =: CG'(\theta)r$ in each coordinate for some constant C (by Assumption (A3)) that depends only on k timesteps, and not n. Subsequently, we suppress vector and matrix notation for the coordinates of θ and its derivatives, and their variances and covariances, with the variance of a vector interpreted as the sum of the variance of its components. We invoke Lemma 2 of Karjalainen et al. (2023) to bound the L_2 error of each of the k (from time $n-k$ to time n) functionals from its expectation under the posterior by $Cr\sqrt{p}G'(\theta)J^{-1/2}k$. That is, we have that

$$\mathbb{E}\left[\|\nabla_\theta \hat{s}_{n,k}^{\alpha}(\theta) - \mathbb{E}_\pi \nabla_\theta \hat{s}_{n,k}^{\alpha}(\theta)\|_2^2\right]^{1/2} \le \frac{Cr\sqrt{p}G'(\theta)k}{\sqrt{J}}. \tag{S193}$$

By the bias-variance decomposition, this in turn bounds the variance at a single timestep by $C^2r^2p\,G'(\theta)^2J^{-1}k^2$. Concretely, we have that

$$\mathrm{Var}\left(\frac{1}{J}\sum_{j=1}^{J}\sum_{i=n-k}^{n} \nabla_\theta \left(\alpha^{n-i}\log\left(g_{i,j}^{A,F,\theta}\right) - \alpha^{n-i+1}\log\left(g_{i-1,j}^{A,F,\theta}\right)\right)\right) \lesssim \frac{r^2k^2p2G'(\theta)^2}{J}. \tag{S194}$$

It now remains to bound the variance of $\nabla_\theta \hat{s}_k^{\alpha}(\theta)$ by considering the covariance of each of the N terms that comprise it. We decompose

$$\begin{aligned}
\mathrm{Var}\big(\nabla_\theta \hat{s}_k^{\alpha}(\theta)\big) &= \mathrm{Var}\left(\sum_{n=1}^{N}\frac{1}{J}\sum_{j=1}^{J}\sum_{i=n-k}^{n} \nabla_\theta \left(\alpha^{n-i}\log\left(g_{i,j}^{A,F,\theta}\right) - \alpha^{n-i+1}\log\left(g_{i-1,j}^{A,F,\theta}\right)\right)\right) \\
&= \sum_{n=1}^{N}\mathrm{Var}\big(\nabla_\theta \hat{s}_{n,k}^{\alpha}(\theta)\big) + 2\sum_{m<n}\mathrm{Cov}\big(\nabla_\theta \hat{s}_{m,k}^{\alpha}(\theta), \nabla_\theta \hat{s}_{n,k}^{\alpha}(\theta)\big) && \text{(S195)} \\
&\lesssim \frac{r^2k^2p\,G'(\theta)^2N}{J} + 2\sum_{m<n}\mathrm{Cov}\big(\nabla_\theta \hat{s}_{m,k}^{\alpha}(\theta), \nabla_\theta \hat{s}_{n,k}^{\alpha}(\theta)\big). && \text{(S196)}
\end{aligned}$$

Similarly to the proof of the MOP-0 case, we use Assumptions (A2) and (A3) that ensure strong mixing. We know from Theorem 3 of Karjalainen et al. (2023) that when $\mathbf{M}_{n,n+k}$ is the k-step Markov operator from timestep n and $\beta_{\mathrm{TV}}(M) = \sup_{x,y\in E}\|M(x,\cdot) - M(y,\cdot)\|_{\mathrm{TV}} = \sup_{\mu,\nu\in\mathcal{P},\mu\neq\nu}\frac{\|\mu M - \nu M\|_{\mathrm{TV}}}{\|\mu-\nu\|_{\mathrm{TV}}}$ is the Dobrushin contraction coefficient of a Markov operator,

$$\beta_{\mathrm{TV}}(\mathbf{M}_{n,n+k}) \le (1-\epsilon)^{\lfloor k/(c\log(J))\rfloor}, \tag{S197}$$

i.e. the mixing time of the particle filter is $O(\log(J))$, where ϵ and c depend on $\bar{M}, \underline{M}, \bar{G}, \underline{G}$ in (A2) and (A3). By Lemma S9, the particle filter itself is strongly mixing, with α-mixing coefficients $a(l) \le (1-\epsilon)^{\lfloor l/(c\log(J))\rfloor}$. Therefore, functions of particles are strongly mixing as well, with α-mixing coefficients bounded by the original (to see this, observe that the σ-algebra of the functionals is contained within the original σ-algebra).

We now derive the α-mixing coefficients of $(\nabla_\theta \hat{s}^\alpha_{n,k}(\theta))_{n=1}^N$. Observe that $\nabla_\theta \hat{s}^\alpha_{n,k}$ is strong mixing at lag $k+1$, as all weights beyond k timesteps are truncated. We therefore have that the mixing time of $\nabla_\theta \hat{s}^\alpha_{n,k}(\theta)$ is $O(1+k+\log(J))$, and that the α-mixing coefficients for $(\nabla_\theta \hat{s}^\alpha_{n,k}(\theta))_{n=1}^N$ are given by $a(l) \leq (1-\epsilon)^{\lfloor l/(c(1+k+\log(J)))\rfloor}$. Therefore, by Davydov's inequality and Lemma 2 of Karjalainen et al. (2023), and noting that $\nabla_\theta \hat{s}^\alpha_{n,k}(\theta) \leq CrG'(\theta)$ by Assumption (A3), by a similar argument to the MOP-0 case, we find that

$$\mathrm{Cov}\big(\nabla_\theta \hat{s}^\alpha_{m,k}(\theta), \nabla_\theta \hat{s}^\alpha_{n,k}(\theta)\big) \lesssim (1-\epsilon)^{\frac{1}{2}\lfloor \frac{|n-m|}{c(1+k+\log(J))}\rfloor} \frac{r^2 p\, G'(\theta)^2}{J}. \tag{S198}$$

Concretely, noting that $\nabla_\theta \hat{s}^\alpha_{n,k}(\theta) \leq CrG'(\theta)$ by Assumption (A3), and, without loss of generality, assuming $\mathbb{E}[(\nabla_\theta \hat{s}^\alpha_{m,k}(\theta))^4]^{1/4} \leq \mathbb{E}[(\nabla_\theta \hat{s}^\alpha_{n,k}(\theta))^4]^{1/4}$, we apply Davydov's inequality to see that

$$\begin{aligned}\mathrm{Cov}\big(\nabla_\theta \hat{s}^\alpha_{m,k}(\theta), \nabla_\theta \hat{s}^\alpha_{n,k}(\theta)\big) &\leq a(n-m)^{1/2}\,\mathbb{E}\big[(\nabla_\theta \hat{s}^\alpha_{m,k}(\theta))^4\big]^{1/4}\,\mathbb{E}\big[(\nabla_\theta \hat{s}^\alpha_{n,k}(\theta))^4\big]^{1/4} && \text{(S199)}\\ &\leq a(n-m)^{1/2}\,\mathbb{E}\big[(\nabla_\theta \hat{s}^\alpha_{n,k}(\theta))^4\big]^{1/2}. && \text{(S200)}\end{aligned}$$

To bound this, we use the fact that

$$\mathbb{E}\big[(\nabla_\theta \hat{s}^\alpha_{n,k}(\theta))^4\big] = \mathbb{E}\Big[\big(\nabla_\theta \hat{s}^\alpha_{n,k}(\theta) - \mathbb{E}\big[\nabla_\theta \hat{s}^\alpha_{n,k}(\theta)\big]\big)^4\Big] + \mathbb{E}\big[(\nabla_\theta \hat{s}^\alpha_{n,k}(\theta))^2\big]^2 \tag{S201}$$

alongside Lemma 2 of Karjalainen et al. (2023), which alongside the fact that

$$\begin{aligned}\mathbb{E}\Big[\big(\nabla_\theta \hat{s}^\alpha_{n,k}(\theta) - \mathbb{E}[\nabla_\theta \hat{s}^\alpha_{n,k}(\theta)]\big)^4\Big] &\lesssim \frac{r^4 p^2 G'(\theta)^4}{J^2}, && \text{(S202)}\\ \mathbb{E}\Big[\big(\nabla_\theta \hat{s}^\alpha_{n,k}(\theta) - \mathbb{E}[\nabla_\theta \hat{s}^\alpha_{n,k}(\theta)]\big)^2\Big] &\lesssim \frac{r^2 p\, G'(\theta)^2}{J}, && \text{(S203)}\end{aligned}$$

allows us to find that

$$\mathbb{E}\Big[\big(\nabla_\theta \hat{s}^\alpha_{n,k}(\theta)\big)^4\Big]^{1/2} \lesssim \sqrt{\frac{r^4 p^2 G'(\theta)^4}{J^2} + \left(\frac{r^2 p\, G'(\theta)^2}{J}\right)^2} = \frac{r^2 p\, G'(\theta)^2}{J}, \tag{S204}$$

and conclude that

$$\mathrm{Cov}\big(\nabla_\theta \hat{s}^\alpha_{m,k}(\theta), \nabla_\theta \hat{s}^\alpha_{n,k}(\theta)\big) \leq (1-\epsilon)^{\frac{1}{2}\lfloor \frac{|n-m|}{c(1+k+\log(J))}\rfloor} \frac{r^2 p\, G'(\theta)^2}{J}. \tag{S205}$$

Putting it all together, we see that

$$\begin{aligned}\mathrm{Var}\big(\nabla_\theta \hat{s}_k(\theta)\big) &= \sum_{n=1}^N \mathrm{Var}\big(\nabla_\theta \hat{s}^\alpha_{n,k}(\theta)\big) + 2\sum_{m<n} \mathrm{Cov}\big(\nabla_\theta \hat{s}^\alpha_{m,k}(\theta), \nabla_\theta \hat{s}^\alpha_{n,k}(\theta)\big) && \text{(S206)}\\ &\leq \frac{Nr^2pk^2G'(\theta)^2}{J} + 2\sum_{n=1}^{N-1}(N-n)(1-\epsilon)^{\frac{1}{2}\lfloor \frac{n}{c(1+k+\log(J))}\rfloor}\frac{r^2p\,G'(\theta)^2}{J} && \text{(S207)}\\ &\leq \frac{r^2pk^2G'(\theta)^2}{J}\left(N + 2\sum_{n=1}^{N-1}(N-n)(1-\epsilon)^{\frac{1}{2}\lfloor \frac{n}{c(1+k+\log(J))}\rfloor}\right) && \text{(S208)}\end{aligned}$$

$$\lesssim \frac{k^2 r^2 p\, G'(\theta)^2 N}{J}. \tag{S209}$$

To continue, define the per-time truncation discrepancy

$$\Delta_{n,k}(\theta) := \nabla_\theta \hat{\ell}_n^\alpha(\theta) - \nabla_\theta \hat{s}_{n,k}^\alpha(\theta),$$

so that

$$\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \hat{s}_k^\alpha(\theta) = \sum_{n=1}^N \Delta_{n,k}(\theta).$$

The pointwise calculation above shows that each coordinate of the tail satisfies

$$\|\Delta_{n,k}(\theta)\|_2 \lesssim \frac{\alpha^k}{1-\alpha}\sqrt{p}\, G'(\theta).$$

However, we do not use the deterministic bound

$$\left\| \sum_{n=1}^N \Delta_{n,k}(\theta) \right\|_2 \le \sum_{n=1}^N \|\Delta_{n,k}(\theta)\|_2,$$

since this would introduce an unnecessary factor of N^2 after squaring.

Instead, we bound the variance of the cumulative truncation error. By Assumptions (A2)–(A3) and the forgetting bound of Karjalainen et al. (2023), the underlying particle process is strongly mixing. Since $\Delta_{n,k}$ is a geometrically weighted tail functional of bounded score increments, the same covariance-summation argument used for the truncated estimator gives

$$|\mathrm{Cov}(\Delta_{m,k}(\theta), \Delta_{n,k}(\theta))| \lesssim \frac{\alpha^{2k}}{(1-\alpha)^2} p G'(\theta)^2\, \rho(|n-m|)$$

for some summable sequence $\rho(\cdot)$, with

$$\sum_{\ell \ge 0} \rho(\ell) \lesssim 1.$$

Consequently,

$$\begin{aligned}
\mathrm{Var}\Big(\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \hat{s}_k^\alpha(\theta)\Big) &= \mathrm{Var}\left(\sum_{n=1}^N \Delta_{n,k}(\theta)\right) \\
&= \sum_{n=1}^N \mathrm{Var}(\Delta_{n,k}(\theta)) + 2\sum_{m<n} \mathrm{Cov}(\Delta_{m,k}(\theta), \Delta_{n,k}(\theta)) \\
&\lesssim N \frac{\alpha^{2k}}{(1-\alpha)^2} p G'(\theta)^2.
\end{aligned}$$

We now compare the variance of the full estimator to the variance of the truncated estimator. For any random vectors X, Y,

$$\mathrm{Var}(Y) \le 2\mathrm{Var}(X) + 2\mathrm{Var}(Y - X).$$

Applying this with

$$Y = \nabla_\theta \hat{\ell}^\alpha(\theta), \qquad X = \nabla_\theta \hat{s}_k^\alpha(\theta),$$

and using the covariance-summation bound above yields

$$\begin{aligned}
\mathrm{Var}\Big(\nabla_\theta \hat{\ell}^\alpha(\theta)\Big) &\le 2\mathrm{Var}(\nabla_\theta \hat{s}_k^\alpha(\theta)) + 2\mathrm{Var}\Big(\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \hat{s}_k^\alpha(\theta)\Big) \\
&\lesssim \frac{k^2 NpG'(\theta)^2}{(1-\alpha)^2 J} + \frac{\alpha^{2k}}{(1-\alpha)^2} NpG'(\theta)^2.
\end{aligned}$$

Since this holds for every $k \le N$, the result follows by minimizing over k.

□

Theorem S5 (MSE and variance bounds for DMOP-α)**.** *Suppose $\alpha \in [0,1)$, $\theta = \phi$, Assumptions (A2), (A3), and (A6) hold, Assumption (A7) in Section S7 holds, and multinomial resampling is used. Then, for a constant independent of N, J, and α,*

$$\begin{aligned}
&\mathbb{E}\left\|\nabla_\theta \ell(\theta) - \nabla_\theta \hat{\ell}^\alpha(\theta)\right\|_2^2 \\
&\quad\lesssim pG'(\theta)^2\left[\frac{N}{J(1-\alpha)^2} + N^2\left\{1-\alpha+\frac{N}{J(1-\alpha)}\right\}^2\right], && \text{(S210)}\\
&\mathrm{Var}\left(\nabla_\theta \hat{\ell}^\alpha(\theta)\right) \lesssim \frac{NpG'(\theta)^2}{J(1-\alpha)^2}. && \text{(S211)}
\end{aligned}$$

Proof. Theorem S6, proved in Section S7, gives the sharper population-bias factor

$$0 \le \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \le \frac{\varrho}{(1-\varrho)^2}(1-\alpha), \qquad \text{(S212)}$$

because $1-\alpha\varrho \ge 1-\varrho$. Substituting Equation (S212) into Equation (S294), while retaining its particle bias term, gives

$$\begin{aligned}
&\mathbb{E}\left\|\nabla_\theta \ell(\theta) - \nabla_\theta \hat{\ell}^\alpha(\theta)\right\|_2^2 \\
&\quad\lesssim pG'(\theta)^2\left[\frac{N}{J(1-\alpha)^2} + N^2\left\{\frac{\varrho}{(1-\varrho)^2}(1-\alpha)+\frac{N}{J(1-\alpha)}\right\}^2\right] \\
&\quad\lesssim pG'(\theta)^2\left[\frac{N}{J(1-\alpha)^2} + N^2\left\{1-\alpha+\frac{N}{J(1-\alpha)}\right\}^2\right], && \text{(S213)}
\end{aligned}$$

which is Equation (S210). Lemma S13 gives Equation (S211) directly. □

S7 Sharper Bias-Variance Analysis

We now establish the estimates used in Theorem S5. Write $\nabla_\theta \ell_n^\alpha(\theta)$ for the nth conditional score produced by the deterministic differentiated population recursion. At $\theta = \phi$, its $\alpha = 1$ version is the exact conditional score $\nabla_\theta \ell_n(\theta)$. A differentiated prediction particle at time m is the pair consisting of $X_{m,j}^{P,\phi}$ and $\nabla_\theta \log w_{m,j}^{P,\theta}\Big|_{\theta=\phi}$.

Assumption (A6) (First-Order Model Regularity)**.** *For every fixed collection of simulator random numbers, the simulated state path and the log measurement densities are continuously differentiable in a neighborhood of ϕ. Differentiation may be interchanged with the expectations defining the filtering recursion. The derivative bound in Assumption (A3) is understood as the total pathwise derivative used by DMOP: uniformly in n, j,*

$$\left\| \nabla_\theta \log g_{n,j}^{A,F,\theta} \Big|_{\theta=\phi} \right\|_\infty \leq B_1 G'(\theta). \tag{S214}$$

We first give sufficient conditions for the two requirements in (A6). Suppose a simulator step can be written as

$$X_n = \texttt{process}_n(X_{n-1}, U_n; \theta), \tag{S215}$$

where the distribution of U_n does not depend on θ. For a differentiable test function q_θ whose derivative is dominated by an integrable function, differentiation under the integral and the chain rule give

$$\begin{aligned}
&\nabla_\theta \mathbb{E}\{q_\theta(\texttt{process}_n(x, U_n; \theta))\} \\
&\quad = \nabla_\theta \int q_\theta(\texttt{process}_n(x, u; \theta))\, \nu(du) \\
&\quad = \int \nabla_\theta \{q_\theta(\texttt{process}_n(x, u; \theta))\}\, \nu(du) \\
&\quad = \int [\partial_\theta q_\theta(\texttt{process}_n(x, u; \theta)) + \nabla_x q_\theta(\texttt{process}_n(x, u; \theta)) \nabla_\theta \texttt{process}_n(x, u; \theta)]\, \nu(du) \\
&\quad = \mathbb{E}\left[\partial_\theta q_\theta(X_n) + \nabla_x q_\theta(X_n) \nabla_\theta X_n \mid X_{n-1} = x\right].
\end{aligned} \tag{S216}$$

Applying Equation (S216) successively to the unnormalized filtering numerator and denominator, and then applying the quotient rule, identifies the differentiated population recursion with the conditional expectation of the pathwise derivative used by DMOP. Thus this identification need not be imposed as a separate stability assumption; domination sufficient for the displayed interchange is enough.

It remains to give primitive conditions for the uniform bound in Equation (S214). Suppose that, for every fixed random seed, the simulator Jacobian with respect to its parent state has operator norm at most $\kappa < 1$, its parameter Jacobian and the derivative of the initial simulated state have norm at most B_1, and the state and parameter derivatives of the log measurement density have norm at most $B_1 G'(\theta)$. Differentiating one simulation step gives

$$\begin{aligned}
\left\| \nabla_\theta X_{n,j}^{P,\theta} \right\| &\leq \kappa \left\| \nabla_\theta X_{n-1,j}^{F,\theta} \right\| + B_1 \\
&\leq B_1 \sum_{s=0}^{n} \kappa^s \leq \frac{B_1}{1-\kappa}.
\end{aligned} \tag{S217}$$

The middle inequality follows by induction from the bound on the initial derivative. Multinomial resampling copies the selected state derivative, so the same bound holds after resampling. The measurement-density chain rule then gives

$$\left\| \nabla_\theta \log g_{n,j}^{A,F,\theta} \right\| \leq \left\| \partial_\theta \log f_{Y_n|X_n}(y_n^* \mid X_{n,j}^{F,\theta}; \theta) \right\|$$

$$+ \left\| \nabla_x \log f_{Y_n|X_n}(y_n^* \mid X_{n,j}^{F,\theta}; \theta) \right\| \left\| \nabla_\theta X_{n,j}^{F,\theta} \right\|$$
$$\leq B_1 G'(\theta) \left(1 + \frac{B_1}{1-\kappa} \right). \tag{S218}$$

Thus the stated primitive conditions imply Equation (S214) after enlarging B_1. They are sufficient, not necessary; the proofs below use only Equation (S214).

Assumption (A6) supplies the first-order simulation regularity used below. Geometric population forgetting follows from the backward-kernel contraction under Assumption (A2), and the particle weak-error bound follows from the full-genealogy argument in Lemma S11.

Unrolling the existing filtered-weight recursion at $\theta = \phi$ gives

$$\nabla_\theta \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} = \sum_{r=0}^{n-1} \alpha^r \, \nabla_\theta \log g_{n-r,j}^{A,F,\theta} \Big|_{\theta=\phi}. \tag{S219}$$

For a complete ancestral path, write h_k, only in this section, for the actual vector mark

$$h_k = \nabla_\theta \log g_k^{A,F,\theta} \Big|_{\theta=\phi} \tag{S220}$$

carried on that path. In the path-space notation of Chan et al. (2018), $m(\xi_t)(h_k)$ is its empirical average among the ancestral paths at time t. The manuscript's existing ρ_t is the corresponding population posterior path measure.

We now identify the lag coefficients in the conditional score. At $\theta = \phi$, all weights equal one, so differentiating the two sample averages in the definition of $L_n^{A,\theta,\alpha}$ gives

$$\begin{aligned} \nabla_\theta \hat{\ell}_n^\alpha(\theta) &= \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} \\ &= \frac{1}{J} \sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \frac{\alpha}{J} \sum_{j=1}^{J} \nabla_\theta \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi}, \end{aligned} \tag{S221}$$

where the second equality uses $w_{n,j}^{P,\theta} = (w_{n-1,j}^{F,\theta})^\alpha$. In the MOP-$\alpha$ recursion, the discounted-weight accumulation and process propagation preserve the index j and occur before the time-n resampling step. The resampling index k_j enters only afterward, through

$$w_{n,j}^{F,\theta} = w_{n,k_j}^{P,\theta} \frac{g_{n,k_j}^\theta}{g_{n,k_j}^\phi}. \tag{S222}$$

Consequently, the second average in Equation (S221) is pathwise

$$\frac{1}{J} \sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} = \frac{\alpha}{J} \sum_{j=1}^{J} \nabla_\theta \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi}, \tag{S223}$$

which is the empirical average over the time-$(n-1)$ filtered genealogy. Substituting Equation (S219) into Equation (S221) and changing the index in the second sum give

$$\nabla_\theta \hat{\ell}_n^\alpha(\theta) = \sum_{r=0}^{n-1} \alpha^r m(\xi_n)(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r m(\xi_{n-1})(h_{n-r})$$

$$= m(\xi_n)(h_n) + \sum_{r=1}^{n-1} \alpha^r \{m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r})\}. \tag{S224}$$

Consequently, define

$$\begin{aligned} \Delta^J_{n,0} &= m(\xi_n)(h_n), & \Delta_{n,0} &= \rho_n(h_n), \\ \Delta^J_{n,r} &= m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r}), & \Delta_{n,r} &= \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}), \quad 1 \le r < n. \end{aligned} \tag{S225}$$

The simulator-compatibility and differentiation-under-the-integral clauses of Assumption (A6) identify the right-hand population quantities with the lag coefficients of the deterministic differentiated recursion. Thus

$$\nabla_\theta \ell_n^\alpha(\theta) = \sum_{r=0}^{n-1} \alpha^r \Delta_{n,r}. \tag{S226}$$

Lemma S10 (Lagwise first-derivative forgetting)**.** *Under Assumptions (A2), (A3), and (A6), there are constants $C < \infty$ and $\varrho \in (0,1)$ such that, for $0 \le r < n$,*

$$\|\Delta_{n,r}\|_2 \le C\sqrt{p}\, G'(\theta)\varrho^r. \tag{S227}$$

Proof. We derive the result from the posterior update and the contraction of the posterior backward kernels.

First suppose that $1 \le r < n$, and put $k = n - r$. Fix a coordinate $i \in \{1, \dots, p\}$, and let $h_{k,i}$ be the corresponding coordinate of the mark in Equation (S220). By Equation (S214),

$$|h_{k,i}| \le B_1 G'(\theta). \tag{S228}$$

The Feynman–Kac posterior update from time $n-1$ to time n is

$$\rho_n(dx_{0:n}) = \frac{\rho_{n-1}(dx_{0:n-1})M_n(x_{n-1}, dx_n)G_n(x_n)}{\rho_{n-1}M_n(G_n)}. \tag{S229}$$

Because $h_{k,i}$ is measurable with respect to the path through time $k \le n-1$, integrating Equation (S229) first with respect to x_n gives

$$\rho_n(h_{k,i}) = \frac{\rho_{n-1}\{h_{k,i}M_n(G_n)\}}{\rho_{n-1}M_n(G_n)}, \tag{S230}$$

where, using the manuscript's kernel notation,

$$M_n(G_n)(x_{n-1}) = \int M_n(x_{n-1}, dx_n)G_n(x_n). \tag{S231}$$

The bounds on G_n and the fact that M_n is a Markov kernel imply for every x_{n-1} that

$$M_n(G_n)(x_{n-1}) \ge \underline{G} \int M_n(x_{n-1}, dx_n) = \underline{G}, \tag{S232}$$

$$M_n(G_n)(x_{n-1}) \le \bar{G} \int M_n(x_{n-1}, dx_n) = \bar{G}. \tag{S233}$$

We next control how much information about the path through time k can remain at time $n-1$. For $s \geq 1$, the posterior backward kernel from X_s to X_{s-1}, written only in this proof as B_s, is

$$B_s(x_s, dx_{s-1}) = \frac{\pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s \mid x_{s-1}; \theta)}{\displaystyle\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s \mid u; \theta)}. \tag{S234}$$

Assumption (A2) gives the following minorization for every measurable set A:

$$\begin{aligned} B_s(x_s, A) &= \frac{\displaystyle\int_A \pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s \mid x_{s-1}; \theta)}{\displaystyle\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s \mid u; \theta)} \\ &\geq \frac{\underline{\mathrm{M}}\, \pi_{s-1}(A)}{\bar{M} \int \pi_{s-1}(du)} \\ &= \frac{\underline{\mathrm{M}}}{\bar{M}} \pi_{s-1}(A). \end{aligned} \tag{S235}$$

Put $\epsilon = \underline{\mathrm{M}}/\bar{M}$. If $\epsilon < 1$, Equation (S235) permits the decomposition

$$B_s(x_s, \cdot) = \epsilon \pi_{s-1}(\cdot) + (1-\epsilon) R_s(x_s, \cdot), \tag{S236}$$

where

$$R_s(x_s, A) = \frac{B_s(x_s, A) - \epsilon \pi_{s-1}(A)}{1-\epsilon} \tag{S237}$$

is a Markov kernel. Hence, for every bounded real function f and any x_s, x_s',

$$\begin{aligned} B_s(f)(x_s) - B_s(f)(x_s') &= (1-\epsilon)\{R_s(f)(x_s) - R_s(f)(x_s')\}, \\ \left|B_s(f)(x_s) - B_s(f)(x_s')\right| &\leq (1-\epsilon)\operatorname{osc}(f). \end{aligned} \tag{S238}$$

If $\epsilon = 1$, Equation (S235) implies $B_s(x_s, \cdot) = \pi_{s-1}(\cdot)$, so the left-hand side is zero. Taking the supremum over x_s, x_s' gives

$$\operatorname{osc}\{B_s(f)\} \leq (1-\epsilon)\operatorname{osc}(f). \tag{S239}$$

Define, only in this proof,

$$H_k(x_k) = \mathbb{E}_{\rho_k}\{h_{k,i} \mid X_k = x_k\}. \tag{S240}$$

All auxiliary simulator seeds and pathwise derivatives through time k are measurable with respect to the past through time k. At $\theta = \phi$, conditional on X_k, the future physical states, observations, and simulator seeds are independent of this past. Hence

$$\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_k = x_k\} = \mathbb{E}_{\rho_k}\{h_{k,i} \mid X_k = x_k\} = H_k(x_k). \tag{S241}$$

Repeated conditioning through the backward kernels gives

$$\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = x_{n-1}\} = (B_{n-1} B_{n-2} \cdots B_{k+1} H_k)(x_{n-1}). \tag{S242}$$

There are $n-1-k = r-1$ backward kernels in this display. Moreover, Equation (S228) gives

$$\operatorname{osc}(H_k) \leq 2\|H_k\|_\infty$$

$$\leq 2B_1 G'(\theta). \tag{S243}$$

If $r = 1$, Equation (S242) contains no backward kernel. If $r \geq 2$, applying Equation (S239) successively gives

$$\begin{aligned} &\operatorname{osc}\left[\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = \cdot\}\right] \\ &\quad \leq 2B_1 G'(\theta) \begin{cases} 1, & r = 1, \\ (1-\epsilon)^{r-1}, & r \geq 2. \end{cases} \end{aligned} \tag{S244}$$

To finish, abbreviate the conditional expectation on the left of Equation (S244) by H, only for the next calculation. Since the X_{n-1}-marginal of ρ_{n-1} is π_{n-1}, the tower property gives

$$\rho_{n-1}(h_{k,i}) = \pi_{n-1}(H), \tag{S245}$$

$$\rho_{n-1}\{h_{k,i} M_n(G_n)\} = \pi_{n-1}\{H M_n(G_n)\}. \tag{S246}$$

Substituting Equations (S245)– (S246) into Equation (S230) and then subtracting $\rho_{n-1}(h_{k,i})$ gives the exact covariance identity

$$\begin{aligned} \Delta_{n,r,i} &= \rho_n(h_{k,i}) - \rho_{n-1}(h_{k,i}) \\ &= \frac{\pi_{n-1}\{H M_n(G_n)\} - \pi_{n-1}(H)\pi_{n-1} M_n(G_n)}{\pi_{n-1} M_n(G_n)} \\ &= \frac{\pi_{n-1}[\{H - \pi_{n-1}(H)\}\{M_n(G_n) - \pi_{n-1} M_n(G_n)\}]}{\pi_{n-1} M_n(G_n)}. \end{aligned} \tag{S247}$$

For any probability measure μ and bounded real f,

$$|f(x) - \mu(f)| \leq \operatorname{osc}(f). \tag{S248}$$

Equations (S232)– (S233) also imply

$$\operatorname{osc}\{M_n(G_n)\} \leq \bar{G} - \underline{\mathrm{G}}. \tag{S249}$$

Taking absolute values in Equation (S247), using Equations (S248)– (S249), and then using Equation (S244), yields

$$\begin{aligned} |\Delta_{n,r,i}| &\leq \frac{\operatorname{osc}(H) \operatorname{osc}\{M_n(G_n)\}}{\underline{\mathrm{G}}} \\ &\leq \frac{2B_1(\bar{G} - \underline{\mathrm{G}})}{\underline{\mathrm{G}}} G'(\theta) \begin{cases} 1, & r = 1, \\ (1-\epsilon)^{r-1}, & r \geq 2. \end{cases} \end{aligned} \tag{S250}$$

Choose any $\varrho \in (1-\epsilon, 1)$. Since $r \geq 1$,

$$\begin{cases} 1, & r = 1, \\ (1-\epsilon)^{r-1}, & r \geq 2 \end{cases} \leq \varrho^{r-1} = \varrho^{-1}\varrho^{r}. \tag{S251}$$

Absorbing the fixed factor ϱ^{-1} into C, summing the squared coordinate bounds, and taking square roots give

$$\|\Delta_{n,r}\|_2 = \left\{ \sum_{i=1}^{p} |\Delta_{n,r,i}|^2 \right\}^{1/2}$$

$$\leq C\sqrt{p}\, G'(\theta)\varrho^r, \qquad 1 \leq r < n. \tag{S252}$$

Finally, when $r = 0$, Equation (S225) and the bounded-mark condition give

$$\begin{aligned}
\|\Delta_{n,0}\|_2 &= \|\rho_n(h_n)\|_2 \\
&\leq \rho_n(\|h_n\|_2) \\
&\leq \sqrt{p}\, B_1 G'(\theta) \\
&\leq C\sqrt{p}\, G'(\theta)\varrho^0.
\end{aligned} \tag{S253}$$

Equations (S252) and (S253) prove Equation (S227).

□

Lemma S11 (Full-genealogy bias of a lag coefficient)**.** *Under Assumptions (A2), (A3), and (A6), with multinomial resampling, there is a constant independent of* n, r, J *such that, for* $0 \leq r < n$*,*

$$\left\|\mathbb{E}\Delta^J_{n,r} - \Delta_{n,r}\right\|_2 \leq \frac{C\sqrt{p}\, G'(\theta)}{J} n. \tag{S254}$$

Proof. At $\theta = \phi$, resampling is multinomial with the ordinary bootstrap filter probabilities, so the particle locations together with their parent indices form the bootstrap particle genealogy. Fix $0 \leq r < n$, put $k = n - r$, and fix a parameter coordinate $i \in \{1, \ldots, p\}$. For any ancestral path through a time $t \geq k$, let $h_{k,i}$ denote the ith coordinate of the existing mark h_k in Equation (S220); as a function on paths through time t, it ignores every coordinate after time k. Thus

$$m(\xi_t)(h_{k,i}) = \frac{1}{J}\sum_{j=1}^{J} h_{k,i}(\xi^j_t), \qquad \rho_t(h_{k,i}) = \int h_{k,i}(x_{0:t})\, \rho_t(dx_{0:t}). \tag{S255}$$

If the simulator stores random seeds or pathwise derivatives, these are included in the ancestral path. Their base law is independent of θ in the reparameterized simulator representation, so, at $\theta = \phi$, they do not alter the transition law of the physical state or the potential G_a. Hence the future normalizer below depends on a path prefix only through its last physical state. Equation (S214) implies, pointwise on every such path,

$$|h_{k,i}(x_{0:t})| \leq \|h_k(x_{0:t})\|_\infty \leq B_1 G'(\theta). \tag{S256}$$

The Feynman–Kac convention of Chan et al. (2018) places in the target at horizon q the potentials through time $q - 1$. To represent the present filtered path law ρ_t, append at horizon $t + 1$ a selection with potential G_t, followed by the identity mutation. The resulting target at horizon $t + 1$ is ρ_t, and, after relabeling the unchanged paths, its path-particle empirical measure is $m(\xi_t)$.

We verify Assumption A1 of Chan et al. (2018) for this augmented model. Let $0 \leq a \leq b \leq t+1$, and let $z = (x_0, \ldots, x_a)$ and $z' = (x'_0, \ldots, x'_a)$ be two path prefixes. If $a = b$, then

$$Q_{a,a}(1)(z) = Q_{a,a}(1)(z') = 1. \tag{S257}$$

If $a < b$ and $a < t$, the Markov property and the definition of the Feynman–Kac semigroup give

$$Q_{a,b}(1)(z) = G_a(x_a) \int f_{X_{a+1}|X_a}(x_{a+1} \mid x_a; \theta) Q_{a+1,b}(1)(x_{a+1})\, dx_{a+1}, \tag{S258}$$

$$Q_{a,b}(1)(z') = G_a(x'_a) \int f_{X_{a+1}|X_a}(x_{a+1} \mid x'_a; \theta) Q_{a+1,b}(1)(x_{a+1})\, dx_{a+1}. \tag{S259}$$

The function $Q_{a+1,b}(1)$ is nonnegative. Assumptions (A2) and (A3) therefore give, term by term,

$$Q_{a,b}(1)(z) \le \bar{G}\bar{M} \int Q_{a+1,b}(1)(x_{a+1})\, dx_{a+1}, \tag{S260}$$

$$Q_{a,b}(1)(z') \ge \underline{\mathrm{GM}} \int Q_{a+1,b}(1)(x_{a+1})\, dx_{a+1}. \tag{S261}$$

The integral in these two displays is the same and is strictly positive. Consequently,

$$Q_{a,b}(1)(z) \le \frac{\bar{M}\bar{G}}{\underline{\mathrm{MG}}} Q_{a,b}(1)(z'). \tag{S262}$$

For the remaining nonzero step, $a = t$ and $b = t+1$, the appended identity mutation gives

$$\begin{aligned} &Q_{t,t+1}(1)(z) = G_t(x_t), \qquad\qquad Q_{t,t+1}(1)(z') = G_t(x'_t), \\ &Q_{t,t+1}(1)(z) \le \frac{\bar{G}}{\underline{\mathrm{G}}} Q_{t,t+1}(1)(z'). \end{aligned} \tag{S263}$$

Together, Equations (S257) and (S262)– (S263) verify Assumption A1 of Chan et al. (2018), uniformly in a, b, t, with a constant no larger than $\bar{M}\bar{G}/(\underline{\mathrm{MG}})$.

By Theorem 2 of Chan et al. (2018), after translating their particle count N to the present particle count J, there is a constant $C < \infty$, depending only on $\bar{M}, \underline{\mathrm{M}}, \bar{G}, \underline{\mathrm{G}}$, such that, whenever $J \ge C(t+1)$, the asserted bound holds for nonnegative path functionals. Applying it to f^+ and f^-, and using $f = f^+ - f^-$, gives, for every integrable real path functional f,

$$|\mathbb{E}m(\xi_t)(f) - \rho_t(f)| \le C\frac{t+1}{J}\rho_t(|f|). \tag{S264}$$

Suppose first that $J \ge Cn$, with C enlarged so that the theorem's condition holds for $t \in \{n-1, n\}$. Since $t+1 \le n+1 \le 2n$ at these two times, substituting $f = h_{k,i}$ in Equation (S264), then using Equation (S256), gives

$$\begin{aligned} |\mathbb{E}m(\xi_t)(h_{k,i}) - \rho_t(h_{k,i})| &\le C\frac{t+1}{J}\rho_t(|h_{k,i}|) \\ &\le C\frac{t+1}{J}\rho_t\{B_1 G'(\theta)\} \\ &= \frac{CB_1G'(\theta)(t+1)}{J} \\ &\le \frac{CG'(\theta)n}{J}, \end{aligned} \tag{S265}$$

where the last line absorbs B_1 into C.

When $r = 0$, we have $k = n$. The first line of Equation (S225) and Equation (S265) with $t = n$ yield

$$\begin{aligned} \left|\mathbb{E}\Delta^J_{n,0,i} - \Delta_{n,0,i}\right| &= |\mathbb{E}m(\xi_n)(h_{n,i}) - \rho_n(h_{n,i})| \\ &\le \frac{CG'(\theta)n}{J}. \end{aligned} \tag{S266}$$

When $1 \le r < n$, we have $k = n - r \le n - 1$, so the same path functional is defined at both times $n-1$ and n. The second line of Equation (S225) gives

$$\begin{aligned}
\mathbb{E}\Delta^J_{n,r,i} - \Delta_{n,r,i} &= \mathbb{E}\{m(\xi_n)(h_{k,i}) - m(\xi_{n-1})(h_{k,i})\} - \{\rho_n(h_{k,i}) - \rho_{n-1}(h_{k,i})\} \\
&= \{\mathbb{E}m(\xi_n)(h_{k,i}) - \rho_n(h_{k,i})\} - \{\mathbb{E}m(\xi_{n-1})(h_{k,i}) - \rho_{n-1}(h_{k,i})\}. \qquad \text{(S267)}
\end{aligned}$$

Taking absolute values, using $|a - b| \le |a| + |b|$, and applying Equation (S265) at the two times gives

$$\begin{aligned}
\left|\mathbb{E}\Delta^J_{n,r,i} - \Delta_{n,r,i}\right| &\le |\mathbb{E}m(\xi_n)(h_{k,i}) - \rho_n(h_{k,i})| + |\mathbb{E}m(\xi_{n-1})(h_{k,i}) - \rho_{n-1}(h_{k,i})| \\
&\le \frac{CB_1G'(\theta)n}{J} + \frac{CB_1G'(\theta)(n-1)}{J} \\
&\le \frac{2CB_1G'(\theta)n}{J} \\
&\le \frac{CG'(\theta)n}{J}, \qquad \text{(S268)}
\end{aligned}$$

where the final line again enlarges C. Combining Equations (S266) and (S268) give

$$\begin{aligned}
\left\|\mathbb{E}\Delta^J_{n,r} - \Delta_{n,r}\right\|_2 &\le \left\{\sum_{i=1}^{p}\left(\frac{CG'(\theta)n}{J}\right)^2\right\}^{1/2} \\
&= \frac{C\sqrt{p}\,G'(\theta)n}{J}. \qquad \text{(S269)}
\end{aligned}$$

It remains to remove the temporary restriction $J \ge Cn$. If $J < Cn$, Equation (S256) implies for every admissible t that

$$|m(\xi_t)(h_{k,i})| \le m(\xi_t)(|h_{k,i}|) \le B_1G'(\theta), \qquad \text{(S270)}$$

$$|\rho_t(h_{k,i})| \le \rho_t(|h_{k,i}|) \le B_1G'(\theta). \qquad \text{(S271)}$$

Thus, for $r = 0$,

$$\left|\mathbb{E}\Delta^J_{n,0,i} - \Delta_{n,0,i}\right| \le 2B_1G'(\theta), \qquad \text{(S272)}$$

whereas, for $1 \le r < n$, Equation (S267) gives

$$\left|\mathbb{E}\Delta^J_{n,r,i} - \Delta_{n,r,i}\right| \le 4B_1G'(\theta). \qquad \text{(S273)}$$

The latter bound is valid in both cases. Therefore

$$\begin{aligned}
\left\|\mathbb{E}\Delta^J_{n,r} - \Delta_{n,r}\right\|_2 &\le 4B_1\sqrt{p}\,G'(\theta) \\
&< 4CB_1\sqrt{p}\,G'(\theta)\frac{n}{J} \\
&\le \frac{C\sqrt{p}\,G'(\theta)n}{J}, \qquad \text{(S274)}
\end{aligned}$$

after one final enlargement of C. This proves Equation (S254) for all $J \ge 1$. $\square$

Lemma S12 (Particle bias of a discounted conditional score)**.** *Under the conditions of Lemma S11, for $\alpha \in [0,1)$,*

$$\left\|\mathbb{E}\nabla_\theta \hat{\ell}_n^\alpha(\theta) - \nabla_\theta \ell_n^\alpha(\theta)\right\|_2 \leq \frac{C\sqrt{p}\,G'(\theta)n}{J(1-\alpha)}. \tag{S275}$$

At $\alpha = 1$, the same argument gives

$$\left\|\mathbb{E}\nabla_\theta \hat{\ell}_n^1(\theta) - \nabla_\theta \ell_n^1(\theta)\right\|_2 \leq \frac{C\sqrt{p}\,G'(\theta)n^2}{J}. \tag{S276}$$

Proof. At $\theta = \phi$, the particles and resampling indices do not depend on α. Taking expectations in Equation (S224), subtracting Equation (S226), and using Equation (S225) give

$$\begin{aligned}
&\mathbb{E}\nabla_\theta \hat{\ell}_n^\alpha(\theta) - \nabla_\theta \ell_n^\alpha(\theta) \\
&\qquad = \sum_{r=0}^{n-1} \alpha^r \left(\mathbb{E}\Delta_{n,r}^J - \Delta_{n,r}\right).
\end{aligned} \tag{S277}$$

Using Lemma S11 and the triangle inequality,

$$\begin{aligned}
\left\|\mathbb{E}\nabla_\theta \hat{\ell}_n^\alpha(\theta) - \nabla_\theta \ell_n^\alpha(\theta)\right\|_2 &\leq \frac{C\sqrt{p}\,G'(\theta)n}{J} \sum_{r=0}^{n-1} \alpha^r \\
&\leq \frac{C\sqrt{p}\,G'(\theta)n}{J} \sum_{r=0}^{\infty} \alpha^r \\
&= \frac{C\sqrt{p}\,G'(\theta)n}{J(1-\alpha)},
\end{aligned} \tag{S278}$$

which proves Equation (S275). When $\alpha = 1$, the same decomposition gives

$$\begin{aligned}
\left\|\mathbb{E}\nabla_\theta \hat{\ell}_n^1(\theta) - \nabla_\theta \ell_n^1(\theta)\right\|_2 &\leq \sum_{r=0}^{n-1} \left\|\mathbb{E}\Delta_{n,r}^J - \Delta_{n,r}\right\|_2 \\
&\leq n\frac{C\sqrt{p}\,G'(\theta)n}{J} = \frac{C\sqrt{p}\,G'(\theta)n^2}{J},
\end{aligned} \tag{S279}$$

which proves Equation (S276). □

Assumption (A7) (Finite-Particle Continuation Stability)**.** *For a coordinate i, let $Z_i = \partial_{\theta_i}\hat{\ell}^\alpha(\theta)$, let $\mathcal{F}_m^J$ contain all differentiated particles through time m, and put $M_{m,i} = \mathbb{E}(Z_i \mid \mathcal{F}_m^J)$. Conditionally on $\mathcal{F}_{m-1}^J$, replace differentiated offspring j at time m by an independent copy from its conditional law, integrate out all later particle randomness, and denote the resulting conditional expectation by $M_{m,i}^{(j)}$. There is a constant independent of $N, J, m, i,$ and $\alpha \in [0,1)$ such that*

$$\left\{\mathbb{E}\left[(M_{m,i} - M_{m,i}^{(j)})^2 \mid \mathcal{F}_{m-1}^J\right]\right\}^{1/2} \leq \frac{CG'(\theta)}{J(1-\alpha)}. \tag{S280}$$

Assumption (A7) concerns the finite-J, $O(J)$ interacting genealogy. Lemmas S10– S12 derive deterministic forgetting, full-genealogy weak bias, and discounted particle bias from the primitive model conditions. The time-uniform results of Del Moral et al. (2015); Tadić and Doucet (2021) establish error and stability bounds for their $O(J^2)$ backward derivative estimator; they do not imply (A7) for DMOP. Assumption (A7) is used only for the variance bound below.

Lemma S13 (Variance of the discounted genealogical score)**.** *Under the conditions of Lemma S12 and Assumption (A7), for* $\alpha \in [0, 1)$*,*

$$\mathrm{Var}\left(\nabla_\theta \hat{\ell}^\alpha(\theta)\right) \lesssim \frac{NpG'(\theta)^2}{J(1-\alpha)^2}, \tag{S281}$$

where $\mathrm{Var}(Z) = \mathbb{E}\|Z - \mathbb{E}Z\|_2^2$ *for a random vector* Z*.*

Proof. We prove the result coordinatewise. Fix i and write $Z_i = \partial_{\theta_i} \hat{\ell}^\alpha(\theta)$. Let $\mathcal{F}_m^J$ now include all differentiated particles through time m, and put $\mathcal{F}_{-1}^J = \{\varnothing, \Omega\}$. The Doob martingale differences

$$D_{m,i} = \mathbb{E}(Z_i \mid \mathcal{F}_m^J) - \mathbb{E}(Z_i \mid \mathcal{F}_{m-1}^J), \qquad m = 0, \dots, N, \tag{S282}$$

are orthogonal in L^2, and their sum is $Z_i - \mathbb{E}Z_i$. To verify these claims, first note that $\mathcal{F}_N^J$ contains all randomness used to construct Z_i, so

$$\begin{aligned} \sum_{m=0}^N D_{m,i} &= \mathbb{E}(Z_i \mid \mathcal{F}_N^J) - \mathbb{E}(Z_i \mid \mathcal{F}_{-1}^J) \\ &= Z_i - \mathbb{E}Z_i. \end{aligned} \tag{S283}$$

If $l < m$, then $D_{l,i}$ is $\mathcal{F}_{m-1}^J$-measurable and

$$\begin{aligned} \mathbb{E}(D_{l,i} D_{m,i}) &= \mathbb{E}\left[D_{l,i} \mathbb{E}(D_{m,i} \mid \mathcal{F}_{m-1}^J)\right] \\ &= 0. \end{aligned} \tag{S284}$$

Squaring Equation (S283), taking expectations, and using Equation (S284) give

$$\mathrm{Var}(Z_i) = \sum_{m=0}^N \mathbb{E}(D_{m,i}^2). \tag{S285}$$

The tower property gives

$$\mathbb{E}(M_{m,i} \mid \mathcal{F}_{m-1}^J) = \mathbb{E}(Z_i \mid \mathcal{F}_{m-1}^J) = M_{m-1,i}. \tag{S286}$$

Consequently,

$$\begin{aligned} \mathbb{E}(D_{m,i}^2 \mid \mathcal{F}_{m-1}^J) &= \mathbb{E}\{(M_{m,i} - M_{m-1,i})^2 \mid \mathcal{F}_{m-1}^J\} \\ &= \mathrm{Var}(M_{m,i} \mid \mathcal{F}_{m-1}^J). \end{aligned} \tag{S287}$$

Conditional on $\mathcal{F}_{m-1}^J$, $M_{m,i}$ is a function of the J conditionally independent offspring at time m. The conditional Efron–Stein inequality, followed by Assumption (A7), therefore gives

$$\begin{aligned} \mathbb{E}(D_{m,i}^2 \mid \mathcal{F}_{m-1}^J) &\le \frac{1}{2} \sum_{j=1}^J \mathbb{E}\{(M_{m,i} - M_{m,i}^{(j)})^2 \mid \mathcal{F}_{m-1}^J\} \\ &\le \frac{J}{2} \left\{\frac{CG'(\theta)}{J(1-\alpha)}\right\}^2 \end{aligned}$$

$$\leq \frac{CG'(\theta)^2}{J(1-\alpha)^2}. \tag{S288}$$

Taking expectations in Equation (S288) and substituting into Equation (S285) give, for $N \geq 1$,

$$\begin{aligned}
\mathrm{Var}(Z_i) &\leq \sum_{m=0}^{N} \frac{CG'(\theta)^2}{J(1-\alpha)^2} \\
&= \frac{C(N+1)G'(\theta)^2}{J(1-\alpha)^2} \\
&\leq \frac{CNG'(\theta)^2}{J(1-\alpha)^2}.
\end{aligned} \tag{S289}$$

Finally,

$$\begin{aligned}
\mathrm{Var}\{\nabla_\theta \hat{\ell}^\alpha(\theta)\} &= \mathbb{E}\left[\sum_{i=1}^{p}(Z_i - \mathbb{E}Z_i)^2\right] \\
&= \sum_{i=1}^{p} \mathrm{Var}(Z_i) \\
&\leq \frac{CNpG'(\theta)^2}{J(1-\alpha)^2},
\end{aligned} \tag{S290}$$

which proves Equation (S281). □

Lemma S14 (Population discounting bias)**.** *Under the conditions of Lemma S10, suppose $\theta = \phi$ and $\alpha \in [0,1]$. Then*

$$\|\nabla_\theta \ell_n^\alpha(\theta) - \nabla_\theta \ell_n(\theta)\|_2 \leq C\sqrt{p}\, G'(\theta) \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \tag{S291}$$

Proof. Equation (S226) at α and at $\alpha = 1$ gives

$$\begin{aligned}
\nabla_\theta \ell_n(\theta) - \nabla_\theta \ell_n^\alpha(\theta) &= \sum_{r=0}^{n-1}(1-\alpha^r)\Delta_{n,r} \\
&= \sum_{r=1}^{n-1}(1-\alpha^r)\Delta_{n,r},
\end{aligned} \tag{S292}$$

because $1 - \alpha^0 = 0$. Taking norms, applying the triangle inequality and Lemma S10, and then extending the nonnegative scalar sum to infinity give

$$\begin{aligned}
\|\nabla_\theta \ell_n(\theta) - \nabla_\theta \ell_n^\alpha(\theta)\|_2 &\leq \sum_{r=1}^{n-1}(1-\alpha^r)\|\Delta_{n,r}\|_2 \\
&\leq C\sqrt{p}\, G'(\theta) \sum_{r=1}^{n-1}(1-\alpha^r)\varrho^r
\end{aligned}$$

$$
\begin{aligned}
&\le C\sqrt{p}\,G'(\theta)\sum_{r=1}^{\infty}(1-\alpha^r)\varrho^r \\
&= C\sqrt{p}\,G'(\theta)\left\{\sum_{r=1}^{\infty}\varrho^r - \sum_{r=1}^{\infty}(\alpha\varrho)^r\right\} \\
&= C\sqrt{p}\,G'(\theta)\left\{\frac{\varrho}{1-\varrho} - \frac{\alpha\varrho}{1-\alpha\varrho}\right\} \\
&= C\sqrt{p}\,G'(\theta)\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \qquad \text{(S293)}
\end{aligned}
$$

□

Theorem S6 (Refined MSE bound for DMOP-α)**.** *Under the conditions of Lemma S12 and Assumption (A7), for $\alpha \in [0,1)$ and $\theta = \phi$,*

$$
\begin{aligned}
&\mathbb{E}\left\|\nabla_\theta\hat{\ell}^\alpha(\theta) - \nabla_\theta\ell(\theta)\right\|_2^2 \\
&\quad\lesssim pG'(\theta)^2\left[\frac{N}{J(1-\alpha)^2} + N^2\left\{\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} + \frac{N}{J(1-\alpha)}\right\}^2\right]. \qquad \text{(S294)}
\end{aligned}
$$

Proof. For a square-integrable random vector Z and a deterministic vector z,

$$
\mathbb{E}\|Z-z\|_2^2 = \mathbb{E}\|Z-\mathbb{E}Z\|_2^2 + \|\mathbb{E}Z - z\|_2^2. \qquad \text{(S295)}
$$

The first term is bounded by Lemma S13. For the second term, use

$$
\nabla_\theta\hat{\ell}^\alpha(\theta) = \sum_{n=1}^{N}\nabla_\theta\hat{\ell}_n^\alpha(\theta), \qquad \nabla_\theta\ell(\theta) = \sum_{n=1}^{N}\nabla_\theta\ell_n(\theta), \qquad \text{(S296)}
$$

add and subtract each population DMOP-α conditional score, and apply the triangle inequality. Lemmas S12 and S14 then give

$$
\begin{aligned}
&\left\|\mathbb{E}\nabla_\theta\hat{\ell}^\alpha(\theta) - \nabla_\theta\ell(\theta)\right\|_2 \\
&= \left\|\sum_{n=1}^{N}\left\{\mathbb{E}\nabla_\theta\hat{\ell}_n^\alpha(\theta) - \nabla_\theta\ell_n^\alpha(\theta)\right\} + \sum_{n=1}^{N}\{\nabla_\theta\ell_n^\alpha(\theta) - \nabla_\theta\ell_n(\theta)\}\right\|_2 \\
&\le \sum_{n=1}^{N}\left\|\mathbb{E}\nabla_\theta\hat{\ell}_n^\alpha(\theta) - \nabla_\theta\ell_n^\alpha(\theta)\right\|_2 \\
&\quad + \sum_{n=1}^{N}\|\nabla_\theta\ell_n^\alpha(\theta) - \nabla_\theta\ell_n(\theta)\|_2 \\
&\le \frac{C\sqrt{p}\,G'(\theta)}{J(1-\alpha)}\sum_{n=1}^{N}n + CN\sqrt{p}\,G'(\theta)\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \\
&= \frac{C\sqrt{p}\,G'(\theta)}{J(1-\alpha)}\frac{N(N+1)}{2} + CN\sqrt{p}\,G'(\theta)\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}
\end{aligned}
$$

$$\leq CN\sqrt{p}\,G'(\theta)\left\{\frac{N}{J(1-\alpha)}+\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}\right\}, \tag{S297}$$

where the equality uses $\sum_{n=1}^{N} n = N(N+1)/2$, and the last line uses $N+1 \leq 2N$. Squaring this bound and using Equation (S295) proves the theorem. □

To make the resulting tradeoff explicit, put $\delta = 1-\alpha$. Since $1-\alpha\varrho = 1-\varrho+\delta\varrho \geq 1-\varrho$,

$$\frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \leq \frac{\varrho}{(1-\varrho)^2}\delta \lesssim \delta. \tag{S298}$$

Theorem S6 and Equation (S298) therefore give

$$\begin{aligned}\mathrm{MSE}_\alpha &\lesssim pG'(\theta)^2\left[\frac{N}{J\delta^2}+N^2\left\{\delta+\frac{N}{J\delta}\right\}^2\right]\\ &\leq CpG'(\theta)^2\left\{\frac{N}{J\delta^2}+N^2\delta^2+\frac{N^4}{J^2\delta^2}\right\},\end{aligned} \tag{S299}$$

where the last inequality uses $(a+b)^2 \leq 2a^2+2b^2$. Here MSE_α denotes the expectation on the left of Equation (S294). If $J \geq N$, the choice

$$\delta = \left(\frac{N}{J}\right)^{1/2} \tag{S300}$$

lies in $[0,1]$, balances the two squared-bias terms, and yields

$$\mathrm{MSE}_\alpha \lesssim pG'(\theta)^2\left(1+\frac{N^3}{J}\right). \tag{S301}$$

Thus the proved upper bound vanishes relative to an N^2 DMOP-0 lower bound when $J/N \to \infty$. It does not, by itself, prove an improvement over an N^2/J DMOP-1 rate.

Corollary S1 (Conditional comparison with the endpoint estimators)**.** *Suppose the conditions of Theorem S6 hold along a joint sequence $N, J \to \infty$, with*

$$\sup_{N,J} pG'(\theta)^2 < \infty. \tag{S302}$$

This condition holds, in particular, when p and θ are fixed. For $a \in \{0,\alpha,1\}$, define all three endpoint and interior errors against the same full score by

$$\mathrm{MSE}_a = \mathbb{E}\left\|\nabla_\theta\hat{\ell}^a(\theta)-\nabla_\theta\ell(\theta)\right\|_2^2. \tag{S303}$$

Suppose, in addition, that the integrated particle genealogy satisfies

$$\left\|\mathbb{E}\Delta_{n,r}^J-\Delta_{n,r}\right\|_2 \leq \frac{C\sqrt{p}\,G'(\theta)}{J}\varrho^r \tag{S304}$$

uniformly in n, r, J, and that there are constants $c_0, c_1 > 0$ such that along this sequence

$$\mathrm{MSE}_0 \geq c_0N^2, \qquad \mathrm{MSE}_1 \geq c_1\frac{N^2}{J}. \tag{S305}$$

If $1 \ll J \ll N$ *and* $1-\alpha = (NJ)^{-1/4}$*, then*

$$\mathrm{RMSE}_\alpha = o(\mathrm{RMSE}_0), \qquad \mathrm{RMSE}_\alpha = o(\mathrm{RMSE}_1), \tag{S306}$$

where $\mathrm{RMSE}_a = \mathrm{MSE}_a^{1/2}$.

Proof. Equation (S304) and the lag decomposition in Equation (S277) give

$$\begin{aligned}&\left\|\mathbb{E}\nabla_\theta \hat{\ell}_n^\alpha(\theta) - \nabla_\theta \ell_n^\alpha(\theta)\right\|_2 \\ &\quad \le \frac{C\sqrt{p}\,G'(\theta)}{J}\sum_{r=0}^{n-1}(\alpha\varrho)^r \le \frac{C\sqrt{p}\,G'(\theta)}{J(1-\varrho)}.\end{aligned} \tag{S307}$$

Repeating the proof of Theorem S6 with this sharper particle-bias bound gives

$$\mathrm{MSE}_\alpha \lesssim pG'(\theta)^2\left\{\frac{N}{J\delta^2} + N^2\delta^2 + \frac{N^2}{J^2}\right\}. \tag{S308}$$

With $\delta = (NJ)^{-1/4}$, the right side of Equation (S308) is at most

$$CpG'(\theta)^2\left(\frac{N^{3/2}}{\sqrt{J}} + \frac{N^2}{J^2}\right). \tag{S309}$$

Relative to the second lower bound in Equation (S305), the two terms in Equation (S309) have ratios $\sqrt{J/N}$ and $1/J$. Relative to the first lower bound, their ratios are $1/\sqrt{NJ}$ and $1/J^2$. All four ratios converge to zero when $1 \ll J \ll N$. Taking square roots proves the result. ☐

Equation (S304) is not a consequence of Assumption (A6). The deterministic filter-derivative stability theorem of Tadić and Doucet (2020) and the time-uniform results for the $O(J^2)$ backward derivative estimator (Del Moral et al., 2015; Tadić and Doucet, 2021) do not establish it for the $O(J)$ DMOP genealogy. The finite-horizon result of Chan et al. (2018) gives the factor n/J in Lemma S11. It cannot be applied only to the last $r+1$ steps of the actual DMOP coefficient because the total reparameterized-simulator derivative carried at time $n-r$ already depends on the earlier genealogy. In particular, that result does not give the geometrically decreasing factor in Equation (S304). Consequently, the corollary records precisely the additional particle-level estimate needed for the sharper regime; it is not claimed under (A6)–(A7) alone.

The two endpoint lower bounds in Equation (S305) are assumptions on the joint sequence, not consequences of the upper bound in Theorem S6. The first formalizes the absence of cancellation in the DMOP-0 population bias. The quadratic lower bound on the path estimator's asymptotic variance in Poyiadjis et al. (2011) motivates the second, but their fixed-horizon central limit theorem does not by itself establish the uniform finite-J lower bound used here.

Remark 1 (Oscillation normalization and the path-space factor)**.** *Suppose a current-state particle theorem gives*

$$\left\|\eta_N^J(h) - \eta_N(h)\right\|_{L^2} \le \frac{C\,\mathrm{osc}(h)}{\sqrt{J}}. \tag{S310}$$

For any current-state test function h_N *with* $\mathrm{osc}(h_N) = O(N)$*, homogeneity gives RMSE* $O(N/\sqrt{J})$ *and MSE* $O(N^2/J)$*. Thus removing an* $\mathrm{osc}(h) \le 1$ *normalization contributes one factor* N^2 *to the MSE, not two.*

A path-space theorem can contain a separate horizon factor. For example, if $\widehat{H}_N$ *estimates the exact path expectation of* H_N*, it may only give*

$$\mathrm{RMSE}(\widehat{H}_N) \leq \frac{CN\,\mathrm{osc}(H_N)}{\sqrt{J}}. \tag{S311}$$

Substituting $\mathrm{osc}(H_N) = O(N)$ *gives RMSE* $O(N^2/\sqrt{J})$ *and MSE* $O(N^4/J)$*. One factor* N *in RMSE therefore comes from the scale of the unnormalized additive score and the other from the path-space stability constant. The quadratic lower bound of Poyiadjis et al. (2011) does not assert that this quartic upper bound is sharp.*

S8 Figures for Bayesian Inference

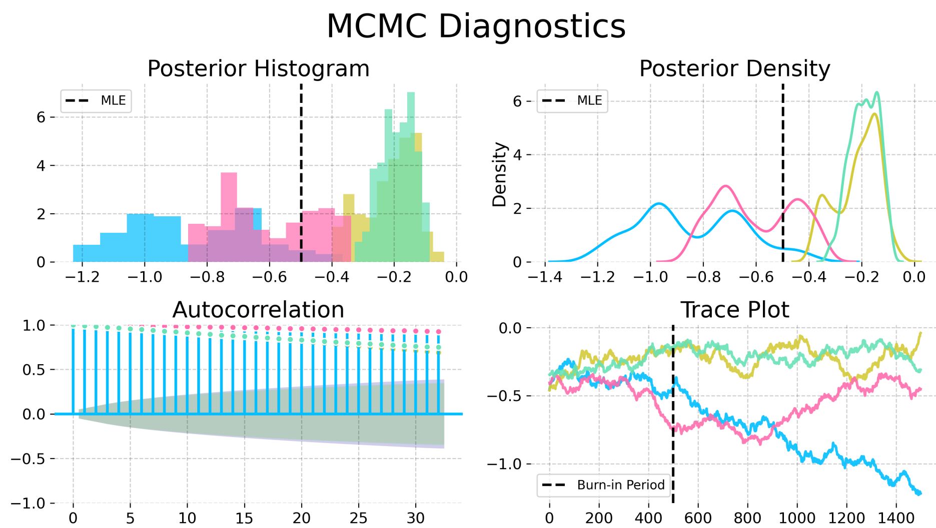

Figure S-1: Convergence diagnostics for the Metropolis-Hastings variant particle MCMC with the random walk proposal and an informative empirical prior from IF2, on the trend parameter in the Dhaka cholera model of King et al. (2008). The poor convergence shown here, using an MCMC algorithm that does not have access to derivatives, contrasts with Fig. 6 in the main text.

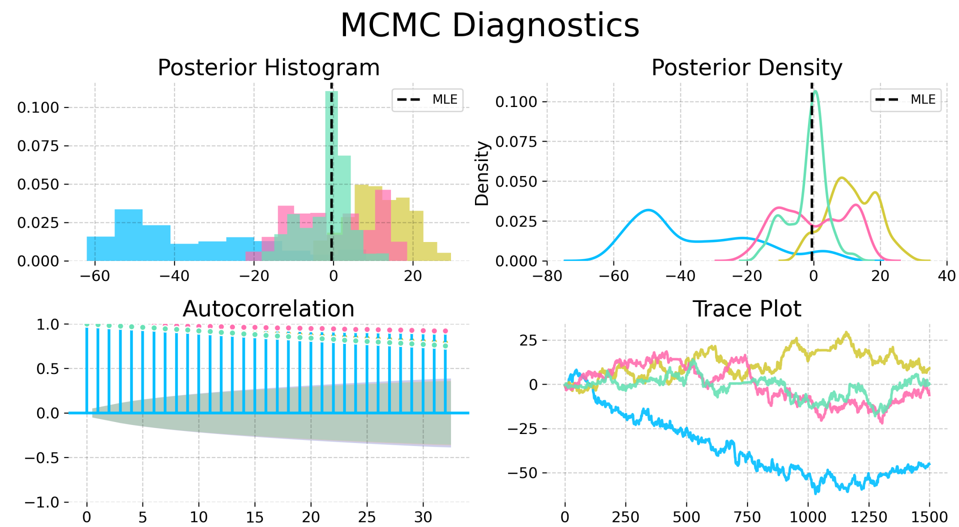

Figure S-2: Convergence diagnostics for a No-U-Turn Sampler (NUTS) with uniform priors on a compact set, on the trend parameter in the Dhaka cholera model of King et al. (2008). The NUTS sampler explores more of the posterior than the Metropolis-Hastings sampler, but fails to converge quickly. This poor convergence, compared with Fig. 6 in the main text, demonstrates the value of the empirical prior for obtaining successful convergence.

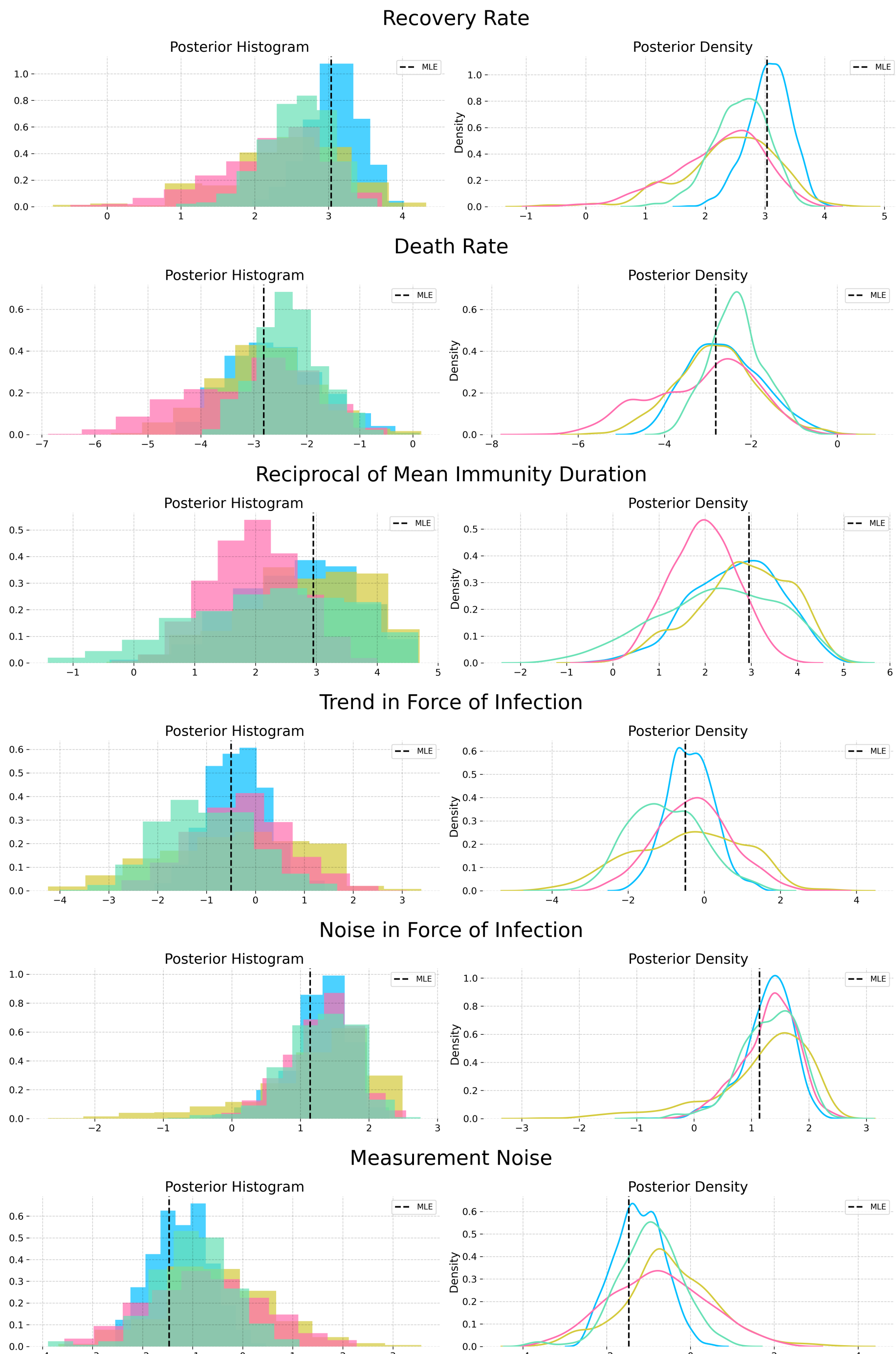

Figure S-3: Posterior estimates from NUTS powered by MOP-α with the informative nonparametric empirical prior obtained from a kernel density estimator on the IF2 parameter cloud. The posterior estimates from each chain are largely in agreement.

S9 Justification for the Choice of Baseline Comparisons

The reasoning we use in the main text to support the magnitude of our contribution is as follows: (i) iterated filtering methods are the current state of the art, and have been for a decade; (ii) we have demonstrated a substantial advance over iterated filtering. The main text focuses on demonstrating (ii), while supporting (i) via references. Listing a few references amounts only to anecdotal evidence, which is not sufficient to persuade someone properly skeptical about the claim. Therefore, this supplement provides an additional quantitative evaluation.

We count the number of papers published in three top scientific journals (*Science*, *Nature*, and *Proceedings of the National Academy of Sciences of the USA (PNAS)*) that carry out statistical inference for partially observed nonlinear non-Gaussian stochastic dynamic models to address a scientific question by fitting mechanistic models to time series data. For all methods, papers in these high-profile journals are expected to represent only be the tip of the iceberg, but this count provides a concrete measure of the capability of the methods to handle world-class scientific inference problems. We could have considered additional journals, but the three we chose are undoubtedly prestigious and their scope covers the full range of scientific endeavor so there is less room for subfield bias. For iterated filtering, our high-impact application count gives the following: 1. He et al. (2023), 2. Fox et al. (2022), 3. Subramanian et al. (2021), 4. Pei et al. (2021), 5. Li et al. (2020), 6. Giezendanner et al. (2020), 7. Wale et al. (2019), 8. Pons-Salort and Grassly (2018), 9. Ranjeva et al. (2017), 10. Martinez et al. (2016) , 11. Becker et al. (2016), 12. Bakker et al. (2016), 13. Laneri et al. (2015), 14. Blake et al. (2014), 15. Blackwood et al. (2013a), 16. Blackwood et al. (2013b), 17. King et al. (2008),

A referee advised consideration of an alternative approach by Gu and Kong (1998) that we found to have a high-impact application count of zero, via Google Scholar.

A referee also recommended comparison with Lindsten et al. (2014), for which the high-impact application count is also zero. Lindsten et al. (2014) has been employed once in *Nature Communications* (Calafat et al., 2018) though that journal did not quite make it onto our elite list. We also found five applications of iterated filtering in *Nature Communications* (Ranjeva et al., 2019; Rodó et al., 2021; Santos-Vega et al., 2022; Briga et al., 2024; Perez-Saez et al., 2025). There is a paper in the Statistics section of *PNAS* by Wang et al. (2022) that demonstrates the method of Lindsten et al. (2014), but here we are counting the demonstrated ability of the methodology to generate high-profile scientific papers, not statistical methodology papers. Otherwise, we would also count Ionides et al. (2006) and Ionides et al. (2015) for iterated filtering.

This reasoning supports iterated filtering as an empirically justified point of reference, in preference to the approaches of Lindsten et al. (2014) and Gu and Kong (1998). Further, the method of Lindsten et al. (2014) does not have the plug-and-play property so it cannot readily be applied to our example. The method of Gu and Kong (1998) would have to be combined with an approach such as Andrieu et al. (2010) that has known scalability problems. Therefore, we focus on comparing with iterated filtering and we do not provide a comparison with Lindsten et al. (2014) or Gu and Kong (1998).

Benchmark tasks are invaluable for making progress on challenging problems (Donoho, 2017). The Dhaka cholera data and model have served as such a benchmark since its introduction by King et al. (2008). Other papers testing methods on this example include Ionides et al. (2015); Fasiolo et al. (2016); Wycoff et al. (2026). Thus, our test is established as a demanding inference problem that challenges the best available methods. We note that this benchmark challenge contains various difficulties that arise commonly in scientific inquiry but are absent from simple SIR models such as the simulation study considered by Lindsten et al. (2014). Ability to accommodate such features is critical for the methodology to be effective for scientific inquiry.

Iterated filtering has recently been used as a benchmark for new POMP model methodology by Whitehouse et al. (2023). This analysis presented numerical results that claimed to beat iterated filtering, but was subsequently found to contain numerical errors. Corrected results are favorable for iterated filtering methods (Hao et al., 2024; Whitehouse et al., 2025; He et al., 2025).

Recent results of Chen et al. (2025) have extended the previous theoretical understanding of iterated filtering. The ongoing study of iterated filtering as a state-of-the-art approach to inference for POMP models gives additional support for our decision to use iterated filtering as a benchmark for our new methodology.

S10 Algorithmic Parameters

We include the full set of algorithmic parameters used for the Dhaka results in Table S-1. The warm start iterations are IF2 iterations, which are followed by gradient descent (GD) iterations when using IFAD. The IF2 cooling rate is the fractional reduction in the parameter perturbations over 50 iterations, so that a cooling rate of 0.5 corresponds to approximately 1% cooling per iteration. The random walk standard deviation (RWSD) scales the noise added at each filtering step to each regular parameter. Initial value parameters (IVPs), that affect the value of the latent process only at time t_0, are treated differently: they are perturbed only at time t_0, with a larger RWSD. Parameters are transformed to a unit scale, for example by taking logarithms or a logistic transformation, so that a single RWSD value works for all parameters.

Hyperparameter	**Standard IF2**	**IFAD-0**	**IFAD-0.97**	**IFAD-1**
Number of Particles (J)	5,000	5,000	5,000	5,000
Warm-start Iterations	650	60	175	60
GD Iterations	—	225	175	225
IF2 Cooling Rate	0.8	0.5	0.5	0.5
Initial RWSD	0.02	0.02	0.02	0.02
Initial RWSD (IVPs)	0.16	0.16	0.16	0.16
Gradient Optimizer	—	Adam	Adam	Adam
Momentum (β_1)	—	0.0	0.9	0.9
LR Schedule	—	Cosine Decay	Cosine Decay	Cosine Decay
Initial LR (η_0)	—	0.1	0.1	0.1
Warm-up Steps	—	10	10	10
Eval. Particles / Reps	5,000 / 36	5,000 / 36	5,000 / 36	5,000 / 36

Table S-1: Algorithmic parameters for the global search benchmarking on the Dhaka cholera model.

Following some experimentation, IFAD was implemnenteed by plugging the DMOP gradient estimate into the Adam optimizer (Kingma and Ba, 2014). The momentum parameter in Adam was turned off for IFAD-0, since this low-variance estimate did not reqquire it. Decreasing the learning rate (LR) was found to be helpful, and we selected the commonly used cosine decay. Warm-up steps were taken with Adam to stabilize its initial performance and avoid noisy

Supplementary References

Andrieu, C., Doucet, A., and Holenstein, R. (2010). Particle Markov chain Monte Carlo methods. *JournaloftheRoyalStatisticalSociety,SeriesB*, 72:269–342.

Bakker, K. M., Martinez-Bakker, M. E., Helm, B., and Stevenson, T. J. (2016). Digital epidemiology reveals global childhood disease seasonality and the effects of immunization. *Proceedings of the National Academy of Sciences*, 113(24):6689–6694.

Becker, A. D., Birger, R. B., Teillant, A., Gastanaduy, P. A., Wallace, G. S., and Grenfell, B. T. (2016). Estimating enhanced prevaccination measles transmission hotspots in the context of cross-scale dynamics. *Proceedings of the National Academy of Sciences*, 113(51):14595–14600.

Blackwood, J. C., Cummings, D. A. T., Broutin, H., Iamsirithaworn, S., and Rohani, P. (2013a). Deciphering the impacts of vaccination and immunity on pertussis epidemiology in Thailand. *ProceedingsoftheNationalAcademyofSciences*, 110:9595–9600.

Blackwood, J. C., Streicker, D. G., Altizer, S., and Rohani, P. (2013b). Resolving the roles of immunity, pathogenesis, and immigration for rabies persistence in vampire bats. *ProceedingsoftheNationalAcademyofSciences*.

Blake, I. M., Martin, R., Goel, A., Khetsuriani, N., Everts, J., Wolff, C., Wassilak, S., Aylward, R. B., and Grassly, N. C. (2014). The role of older children and adults in wild poliovirus transmission. *ProceedingsoftheNationalAcademyofSciences*, 111(29):10604–10609.

Briga, M., Goult, E., Brett, T. S., Rohani, P., and Domenech de Cellès, M. (2024). Maternal pertussis immunization and the blunting of routine vaccine effectiveness: A meta-analysis and modeling study. *Nature Communications*, 15(1):921.

Calafat, F. M., Wahl, T., Lindsten, F., Williams, J., and Frajka-Williams, E. (2018). Coherent modulation of the sea-level annual cycle in the United States by Atlantic Rossby waves. *Nature Communications*, 9(1):2571.

Chan, H. P., Del Moral, P., and Jasra, A. (2018). A sharp first order analysis of Feynman–Kac particle models. *Stochastic Processes and their Applications*, 128(1):332–353.

Chen, Y., Gerber, M., Andrieu, C., and Douc, R. (2025). Self-organizing state-space models with artificial dynamics. *JournaloftheRoyalStatisticalSociety,SeriesB*, page qkaf069.

Chopin, N. (2004). Central limit theorem for sequential Monte Carlo methods and its application to Bayesian inference. *AnnalsofStatistics*, 32:2385–2411.

Chopin, N. and Papaspiliopoulos, O. (2020). *An Introduction to Sequential Monte Carlo*. Springer.

Del Moral, P. (2004). *Feynman-Kac Formulae: Genealogical and Interacting Particle Systems with Applications*. Springer, New York.

Del Moral, P. and Doucet, A. (2003). On a class of genealogical and interacting Metropolis models. In Azéma, J., Émery, M., Ledoux, M., and Yor, M., editors, *Séminaire de Probabilités XXXVII*, pages 415–446. Springer, Berlin.

Del Moral, P., Doucet, A., and Singh, S. S. (2015). Uniform stability of a particle approximation of the optimal filter derivative. *SIAM Journal on Control and Optimization*, 53(3):1278–1304.

Del Moral, P. and Rio, E. (2011). Concentration inequalities for mean field particle models. *The Annals of Applied Probability*, 21(3).

Donoho, D. (2017). 50 years of data science. *Journal of Computational and Graphical Statistics*, 26(4):745–766.

Fasiolo, M., Pya, N., and Wood, S. N. (2016). A comparison of inferential methods for highly nonlinear state space models in ecology and epidemiology. *StatisticalScience*, 31(1):96–118.

Fox, S. J., Lachmann, M., Tec, M., Pasco, R., Woody, S., Du, Z., Wang, X., Ingle, T. A., Javan, E., Dahan, M., Gaither, K., Escott, M. E., Adler, S. I., Johnston, S. C., Scott, J. G., and Meyers, L. A. (2022). Real-time pandemic surveillance using hospital admissions and mobility data. *ProceedingsoftheNationalAcademyofSciences*, 119(7):e2111870119.

Giezendanner, J., Pasetto, D., Perez-Saez, J., Cerrato, C., Viterbi, R., Terzago, S., Palazzi, E., and Rinaldo, A. (2020). Earth and field observations underpin metapopulation dynamics in complex landscapes: Near-term study on carabids. *Proceedings of the National Academy of Sciences*, 117(23):12877–12884.

Gu, M. G. and Kong, F. H. (1998). A stochastic approximation algorithm with Markov chain Monte-Carlo method for incomplete data estimation problems. *Proceedings of the National Academy of Sciences*, 95(13):7270–7274.

Hao, Y., Abkemeier, A. J., and Ionides, E. L. (2024). Poisson approximate likelihood compared to the particle filter. *arXiv:2409.12173*.

He, D., Lin, L., Artzy-Randrup, Y., Demirhan, H., Cowling, B. J., and Stone, L. (2023). Resolving the enigma of Iquitos and Manaus: A modeling analysis of multiple COVID-19 epidemic waves in two Amazonian cities. *Proceedings of the National Academy of Sciences*, 120(10):e2211422120.

He, K., Hao, Y., and Ionides, E. L. (2025). Poisson Approximate Likelihood versus the block particle filter for a spatiotemporal measles model. *arXiv:2507.09121*.

Ionides, E. L., Bretó, C., and King, A. A. (2006). Inference for nonlinear dynamical systems. *ProceedingsoftheNationalAcademyofSciences*, 103:18438–18443.

Ionides, E. L., Nguyen, D., Atchadé, Y., Stoev, S., and King, A. A. (2015). Inference for dynamic and latent variable models via iterated, perturbed Bayes maps. *ProceedingsoftheNationalAcademyofSciences*, 112(3):719—724.

Karjalainen, J., Lee, A., Singh, S. S., and Vihola, M. (2023). On the forgetting of particle filters. *arXiv:2309.08517*.

King, A. A., Ionides, E. L., Pascual, M., and Bouma, M. J. (2008). Inapparent infections and cholera dynamics. *Nature*, 454:877–880.

Kingma, D. P. and Ba, J. (2014). Adam: A method for stochastic optimization. *arXiv:1412.6980*.

Laneri, K., Paul, R. E., Tall, A., Faye, J., Diene-Sarr, F., Sokhna, C., Trape, J.-F., and Rodó, X. (2015). Dynamical malaria models reveal how immunity buffers effect of climate variability. *Proceedings of the National Academy of Sciences*, 112(28):8786–8791.

Li, R., Pei, S., Chen, B., Song, Y., Zhang, T., Yang, W., and Shaman, J. (2020). Substantial undocumented infection facilitates the rapid dissemination of novel coronavirus (SARS-CoV-2). *Science*, 368(6490):489–493.

Lindsten, F., Jordan, M. I., and Schön, T. B. (2014). Particle Gibbs with ancestor sampling. *The Journal of Machine Learning Research*, 15(1):2145–2184.

Martinez, P. P., King, A. A., Yunus, M., Faruque, A., and Pascual, M. (2016). Differential and enhanced response to climate forcing in diarrheal disease due to rotavirus across a megacity of the developing world. *Proceedings of the National Academy of Sciences*, 113(15):4092–4097.

Naesseth, C., Linderman, S., Ranganath, R., and Blei, D. (2018). Variational sequential Monte Carlo. In Storkey, A. and Perez-Cruz, F., editors, *Proceedings of the Twenty-First International Conference on Artificial Intelligence and Statistics*, volume 84 of *Proceedings of Machine Learning Research*, pages 968–977. PMLR.

Pei, S., Liljeros, F., and Shaman, J. (2021). Identifying asymptomatic spreaders of antimicrobial-resistant pathogens in hospital settings. *Proceedings of the National Academy of Sciences*, 118(37):e2111190118.

Perez-Saez, J., Bellon, M., Lessler, J., Berthelot, J., Hodcroft, E. B., Michielin, G., Pennacchio, F., Lamour, J., Laubscher, F., L'Huillier, A. G., Posfay-Barbe, K. M., Maerkl, S. J., Guessous, I., Azman, A. S., Eckerle, I., Stringhini, S., and Lorthe, E. (2025). Evolving infectious disease dynamics shape school-based intervention effectiveness. *Nature Communications*, 16(1):6597.

Pons-Salort, M. and Grassly, N. C. (2018). Serotype-specific immunity explains the incidence of diseases caused by human enteroviruses. *Science*, 361(6404):800–803.

Poyiadjis, G., Doucet, A., and Singh, S. S. (2011). Particle approximations of the score and observed information matrix in state space models with application to parameter estimation. *Biometrika*, 98(1):65–80.

Ranjeva, S., Subramanian, R., Fang, V. J., Leung, G. M., Ip, D. K., Perera, R. A., Peiris, J. M., Cowling, B. J., and Cobey, S. (2019). Age-specific differences in the dynamics of protective immunity to influenza. *Nature Communications*, 10(1):1660.

Ranjeva, S. L., Baskerville, E. B., Dukic, V., Villa, L. L., Lazcano-Ponce, E., Giuliano, A. R., Dwyer, G., and Cobey, S. (2017). Recurring infection with ecologically distinct HPV types can explain high prevalence and diversity. *ProceedingsoftheNationalAcademyofSciences*, 114(51):13573–13578.

Rodó, X., Martinez, P. P., Siraj, A., and Pascual, M. (2021). Malaria trends in Ethiopian highlands track the 2000 'slowdown'in global warming. *Nature Communications*, 12(1):1555.

Roosta-Khorasani, F. and Mahoney, M. W. (2016). Sub-sampled Newton methods I: Globally convergent algorithms. *arXiv:1601.04737*.

Santos-Vega, M., Martinez, P., Vaishnav, K., Kohli, V., Desai, V., Bouma, M., and Pascual, M. (2022). The neglected role of relative humidity in the interannual variability of urban malaria in Indian cities. *Nature Communications*, 13(1):533.

Ścibior, A. and Wood, F. (2021). Differentiable particle filtering without modifying the forward pass. *arXiv:2106.10314*.

Subramanian, R., He, Q., and Pascual, M. (2021). Quantifying asymptomatic infection and transmission of COVID-19 in New York City using observed cases, serology, and testing capacity. *Proceedings of the National Academy of Sciences*, 118(9).

Tadić, V. Z. B. and Doucet, A. (2020). Stability of optimal filter higher-order derivatives. *Stochastic Processes and their Applications*, 130(8):4808–4858.

Tadić, V. Z. B. and Doucet, A. (2021). Bias of particle approximations to optimal filter derivative. *SIAM Journal on Control and Optimization*, 59(1):727–748.

Wale, N., Jones, M. J., Sim, D. G., Read, A. F., and King, A. A. (2019). The contribution of host cell-directed vs. parasite-directed immunity to the disease and dynamics of malaria infections. *Proceedings of the National Academy of Sciences*, 116(44):22386–22392.

Wang, E. T., Vannucci, M., Haneef, Z., Moss, R., Rao, V. R., and Chiang, S. (2022). A Bayesian switching linear dynamical system for estimating seizure chronotypes. *Proceedings of the National Academy of Sciences*, 119(46):e2200822119.

Whitehouse, M., Whiteley, N., and Rimella, L. (2023). Consistent and fast inference in compartmental models of epidemics using Poisson Approximate Likelihoods. *JournaloftheRoyalStatisticalSociety,SeriesB*, 85(4):1173–1203.

Whitehouse, M., Whiteley, N., and Rimella, L. (2025). Correction to: Consistent and fast inference in compartmental models of epidemics using Poisson approximate likelihoods. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, pre-published online.

Wycoff, N., Smith, J. W., Booth, A. S., and Gramacy, R. B. (2026). Voronoi candidates for Bayesian optimization. *Journal of Global Optimization*, 94:175–201.